

Follow the Leader:

Informative Voluntary Disclosure and Exploitation

Tal Fishman*
Department of Economics
Princeton University

November 20, 2006

Abstract

This paper develops a model of strategic disclosure by large traders with private information. Trading on private information comes with the risk that adverse price moves will force liquidation before the information is realized. Voluntary disclosure of holdings lets a well-reputed trader profit from the immediate price reaction, before any potential forced liquidation. However, he can also exploit his reputation by disclosing uninformative liquidity trades to earn additional profits. We analyze this situation in a model of a repeated interaction between the public and an informed trader with a reputation for receiving private informative signals. We show that the trader will optimally restrict his frequency of exploitation, but will engage in more exploitation as his reputation improves. Consequently, statements by top experienced traders may be less informative than those made by younger traders with a strong incentive to build a reputation for the future. We test these and other empirical implications of the model using data on voluntary disclosures by mutual fund managers.

Keywords: Large Trader, Trade Disclosure, Forced Liquidation Risk, Reputation, Repeated Games.

JEL classification: G14

*I am grateful to Roland Benabou, Markus Brunnermeier, Sylvain Champonnois, Martin Oehmke, and Hyun Song Shin, as well as to the participants at the Finance Student Discussion Workshop and the Civitas Foundation Finance Seminar at Princeton University for their comments and suggestions. Harrison Hong and Wei Xiong deserve special mentions for their valuable support and advice. Any errors are mine alone. Comments welcome, please send to tfishman@princeton.edu.

1 Introduction

In this paper, we study the degree to which voluntary disclosures by large traders affects asset prices by revealing private information. One of the most successful traders of all time, famous for moving the markets with his public statements, is George Soros. After nearly single-handedly forcing the Bank of England to abandon the European Exchange Rate Mechanism (and profiting massively from the ensuing depreciation in Sterling), Soros' public reputation skyrocketed. A few months later, in April 1993, Soros purchased two to three million ounces of gold at \$345 and invested \$400 million in Newmont Mining, amounting to a 13% stake. When traders learnt of Soros' stake, gold rose \$5 to over \$350, for first time since October 1992. On August 15 the *Sunday Times* of London reported that Soros had sold his entire gold bullion at between \$385 and \$395 an ounce.

In June 1993 Soros repeated the feat, acquiring a 4.8 percent stake in British Land Co. As Slater (1996) recounts in an autobiography of Soros (pp199-203), "if Soros was buying into British real estate, that had to mean that the real estate market in Great Britain had bottomed out. That, at any rate, was how British investors read the Soros investment. The effect on the stock market values of property firms was earth-shattering: ... The shares of British Land itself rose from 298 pence to 484 pence." In response to these incidents, David C. Roche, a London-based strategist for Morgan Stanley, remarked in a June 10, 1993 article in the *New York Times*, "It's a new way of making money, a combination of judicious investment at the bottom of a market and a publicity coup."¹

Most of the reaction to Soros' disclosures was not immediate, but rather was the catalyst for an ensuing rally lasting a few months. As the anecdotes suggest, Soros' disclosure may have been the marginal factor convincing other investors to participate in a rally that would likely happen anyhow for sound long-run fundamental reasons. In the case of gold, prices remained at elevated levels for quite a while, peaking at about \$405 in early 1996. Hence these anecdotes suggest that voluntary disclosure has the potential to hasten the

¹Traders may also publicize their opinions via small private forums. The Triangle Investment Club, a monthly fund manager meeting where members are expected to talk about their latest trading ideas, is exactly one such forum. Biggs (2006) describes the scene as follows. "There were about 25 guys there, about half from hedge funds and the other half very aggressive long-only managers. ... In no more than four minutes, each guy is supposed to elucidate his favorite stock or concept and the reasons why. ... It is assumed you're going to be recommending what you already have a position in." He also tells of how some guys exploited their reputation in the group by "sandbagging," and how the group punished them in response:

"Sandbagging is when you tell a story and egregiously [exaggerate the fundamentals in your favor], hoping to attract buyers at the same time you are actually secretly selling the stock. It's one thing to promote the story of what you own or are short, and as noted, most members make it clear that they have a position one way or the other in the merchandise they are talking about. It's quite another to sandbag. Sandbagging is dishonest, cheating, a violation of the rules of the money game." ("Hedgehogging," Barton Biggs, pp 2-3)

arrival of a coming rally. The ability to make informative disclosures is potentially quite valuable to a speculator possessing private information² regarding the fundamental value of an asset. If the speculator puts on a trade too early, he may be unable to wait for the coming rally (or decline). An adverse price move, for example, may compel him to liquidate his position immediately. Voluntary disclosure increases awareness regarding the mispricing and encourages others to follow his lead. When others follow him, they bring about a more immediate price reaction, thus reducing the potential loss from early liquidation.

This paper follows a large literature, beginning with DeLong, Shleifer, Summers, and Waldmann (1990), studying traders who face a risk of being forced into early liquidation.³ Dow and Gorton (1994) first proposed that privately informed traders will not act on their information if they are unable to hold the asset until the date when the information is reflected in the price. The inability to hold on to positions may be the equilibrium result of a high cost-of-carry, as first noted by Shleifer and Vishny (1990). Alternatively, a money manager may need to show good performance over a short horizon at the end of which his performance is assessed.⁴ Short holding times may also result from uncertainty about when other traders will learn the common information, as in the paper by Abreu and Brunnermeier (2002). Managers may need to liquidate existing holdings to invest in better opportunities as they arise. Finally, early liquidation could also result from the need to meet margin calls in response to exogenous shocks to other holdings or to trading capital.⁵

A market “leader,” able to influence the movement of markets by his words, clearly benefits from this ability, but it is not obvious whether the market will always heed his advice. The public must carefully evaluate whether any leader truly possesses some informational edge or is merely a *charlatan*. Even if the leader is *genuine*, having some private information at least some of the time, he may still be exploiting his reputation. A leader may employ voluntary disclosure to earn additional profits on uninformative liquidity trades. For example, the leader may have initially entered into a trade based on private information but subsequently discovered countervailing information. By disclosing the unprofitable trade that he intends to eliminate from his portfolio, he may realize a better exit price. Such *deceptive*

²Private information here can also come from a superior ability to process public information.

³In one of the most famous examples of such a forced liquidation, Long-Term Capital Management (LTCM), a highly leveraged hedge fund, had to liquidate their entire holdings of roughly \$125 billion as a result of losses in only a few areas amounting to about \$4.6 billion. Their primary position, an on-the-run/off-the-run government bond arbitrage, subsequently recovered and even yielded a profit, but only after LTCM was wiped out. The case of LTCM makes clear how fear of such liquidations may prevent money managers from acting on long-run information.

⁴See, e.g., Dow and Gorton (1997) and Goldman and Sleazak (2003) for a discussion of how the short evaluation horizons that prevail in the mutual fund industry affect portfolio manager behavior.

⁵See, e.g., Brunnermeier and Pederson (2005) for a discussion of the dangers of forced liquidation.

disclosures add to the leader's profits but wind up being losing trades for the public.

In the two examples involving George Soros, both the leader and his following public benefited from his disclosures. The following anecdote illustrates how leaders make deceptive disclosures for private gain. In early November 1995 Jeffrey Vinik, manager of the Fidelity Magellan fund, told the press that "valuations are reasonable and the fundamentals are still outstanding" on Micron Technology, one of his fund's major holdings. Yet, as the Boston Herald reported on December 12, 1995, "On Oct. 31, Magellan had 10.5 million shares of Micron. By Nov. 30, the fund held less than 1.2 million shares, according to SEC filings. Why not say you're selling if you're selling? Because when you're Jeff Vinik, managing the country's largest and most scrutinized stock fund, *the stock is likely to drop in price merely at the mention you're selling.*" (emphasis added) Micron stock subsequently fell 39% in the year after November 1995.

If a genuine leader has the ability to engage in such exploitation as frequently as he wishes, he may end up exploiting the public far more often than he is releasing true information. Consequently, the public will believe that any given statement is unlikely to be informative and will come to ignore all of the leader's statements. The primary contribution of this paper is to analyze the incentives of the leader and the public and demonstrate that informative voluntary disclosure can be sustained (under certain conditions) in spite of the incentive to exploit reputation. Towards this end, we develop a dynamic reputation model of a repeated interaction between a privately informed trader and the public.

The model defines whether a leader is a charlatan or genuine as his type, and reputation is the public's estimate of the probability of being of the genuine type. The public reevaluates the leader after each of his disclosures and updates his reputation over time based on his record of correct and incorrect predictions. The public recognizes that even a genuine leader will occasionally make uninformative disclosures. Thus the reputation moves slowly towards zero (charlatan for sure) or one (genuine for sure) as a complex Markov process.

We find that the credible transmission of information requires that the leader incur some fixed cost of disclosure. Disclosure costs act as a commitment mechanism for the leader, preventing him from making endless deceptive disclosures. Without such a commitment mechanism, honest disclosures are swamped by deceptive disclosures and nothing the leader says is considered credible. Only if disclosure costs are sufficiently high will the leader be dissuaded from making public statements not based on private information.

When a genuine leader is young, before he has had a chance to build a reputation, market

prices do not react much to his announcements and his ability to profit from private information is limited. In order to improve his reputation, he will refrain from exploiting the public. However, as his reputation solidifies, he will seek to earn greater profits by making more deceptive disclosures. Consequently, statements by experienced traders with excellent track records may be less informative than those made by younger traders with a strong incentive to build a reputation for the future.

The relevance and applicability of such a model cannot be confidently asserted without an empirical investigation into the market impact of public statements. Towards this end, we have assembled a dataset of voluntary public statements by mutual fund managers and matched it to widely available data on mutual fund holdings and abnormal returns on those holdings. We find strong evidence that frequent discussions with the press on the part of a portfolio manager improves the early returns on his most recent acquisitions. Furthermore, the impact of public statements is significantly higher for managers of funds with very good past performance. Our results suggest that voluntary public statements may have an economically meaningful impact. We also present a framework for testing some of the model's novel predictions, such as that portfolio managers with better reputations exploit the public more often.

As a result of the early liquidation risk described above, a trader looking to profit from private information concerning long-run fundamentals faces a dilemma. Act too early, and he may be forced to liquidate before others in the market learn his information. Act too late, and he risks missing the trade altogether. The key insight of Froot, Scharfstein, and Stein (1992) into this dilemma is that if the trader has a choice of what information to pursue, he will attempt to learn information that others will soon learn as well, while relatively obscure information remains sidelined. This paper shows that disclosure may allow certain well-reputed traders to profit from truly private information. Consequently, such traders are also less susceptible to the informational herding Froot, Scharfstein, and Stein (1992) describe.

Our model yields many additional long-run results. For example, the market price reaction to a genuine leader become less efficient over time. In other words, the post-announcement price becomes on average further from the fundamental value incorporating all information, including the leader's private information. Two competing effects make the overall outcome ambiguous. A genuine leader's reputation improves over time, and therefore so too does the price reaction to each of his statements. Since the genuine leader does have

private information regarding fundamental value at least some of the time, price efficiency may be improved. However, better reputation also leads to more deception. Since each deceptive disclosure is uninformative, price efficiency turns out to be impaired. Additional long-run results examine the impact of liquidity, probability of early liquidation, and the strength of information signals.

The remainder of the paper proceeds as follows. The following section briefly reviews the related literature. Section 3 presents our assumptions regarding the various market participants and the structure of trading for a single trading round. We also outline certain aspects of the short-run equilibrium in that section, but details of the derivation are relegated to Appendix A. The dynamic reputation game is solved in section 4. Section 5 presents some long-run results concerning the time path of exploitation and market reactions. In section 6 we describe our framework for testing the model’s empirical implications and present some early results. Section 7 concludes. Appendix B presents complete and proposed extensions of the short-run model. Proofs are in Appendix C.

2 Related Literature

A few other papers study strategic disclosure by traders in various contexts. The one closest to our own is Benabou and Laroque (1992). In fact, the example they use to illustrate the ability of some traders to influence markets by their announcements is equally applicable to our model:

In the nervous market of 1987, Mr. Prechter has emerged as both prophet and deity, an adviser whose advice reaches so many investors that he tends to pull the market the way he has predicted it will move.... Mr. Prechter’s words carry such power because he appears to have called the broad outlines of the bull market right since 1982, although he has made some bad calls on short-term moves [*International Herald Tribune*, October 3, 1987, quoted from Benabou and Laroque (1992)]

Similarly to this paper, Benabou and Laroque examine the credibility of disclosure by an informed trader, whom they call a “journalist,” in a repeated interaction with the public. Both papers examine similar ways in which such traders exploit their reputations. In both cases, traders deliberately mislead the public for trading gains. Both papers also demonstrate that such traders curb their exploitation to maintain credibility in a repeated

setting. Their journalist tells the public the opposite of what he believes in order to obtain a better entry price but will occasionally tell the truth. Likewise, our leader tries to convince the public that he has information when he really does not in order to obtain a better exit price but will often remain silent.

The distinguishing feature of this paper is that both the genuine and charlatan leader types are rational optimizers. Both types of leaders face a choice when uninformed between exploiting the market by disclosing uninformative trades and remaining silent. However, the alternative to the rational “opportunistic” journalist of Benabou and Laroque is a non-optimizing “honest” type that tells the truth despite his incentives to exploit.⁶ The opportunistic journalist is only able to profit because other market participants may mistakenly believe him to be honest. Hence their results only concern situations where the market’s assessment of the key player is wrong. Our more elaborate assumptions allow us to study the optimal behavior of a more realistic “good” type, as well as how the behavior of a “bad” type may change when his decisions affect the actions of the other type.

Another key difference is in the reasons behind honest disclosure. Unlike the interaction studied in this paper, the interests of the journalist and his audience are diametrically opposed. The journalist, who receives a signal in every round, only trades *after* making an announcement. Thus he seeks to manipulate the market by deliberately releasing a message opposite to that of his signal. His credibility depends on occasionally going against his interests and telling the truth. In contrast, the informed trader we study profits from honest disclosures but may earn additional profits by releasing misinformation regarding liquidity trades. Hence we focus on the trader’s long-run ability to disseminate truthful information despite the temptation to deceive, whereas Benabou and Laroque focus on the ability to manipulate markets.

Corsetti, Dasgupta, Morris, and Shin (2004) also examine a case where disclosure by informed traders can affect the markets. However, the context they examine and mechanism they use are markedly different from the context and mechanism of this paper. The situation they adopt is one where speculators attack a currency peg. Speculators are differentially informed as to the vulnerability of the currency to attack, but a successful attack requires that a sufficient number of them believe that it is vulnerable. If a large trader is permitted to signal his attack, other traders who may otherwise have thought the currency peg is strong will instead choose to attack as well. Hence their paper presents a mechanism in which the

⁶Benabou and Laroque emphasize that such behavior may still be rational, either because this type is more likely to get caught (e.g. by the SEC) or because he suffers a higher loss (e.g. in reputation, career, prison time, etc.) when caught.

large trader influences the market independently of his information. In contrast, the leader's influence stems entirely from his information, and it is the evolution of his reputation for possessing such information that is the focus of this paper.

Numerous papers discuss trade disclosure in other contexts. At least three papers study the effect of financial market regulations requiring insiders to disclose their trades on insiders' trading strategies and expected profits. In Fishman and Hagerty (1995), market makers cannot distinguish between uninformed insiders and informed insiders. Thus an uninformed insider may profit from moving the price with an initial trade and unwinding the position in the next round of trading. John and Narayanan (1997) consider the case where insiders are informed for sure and there is an asymmetry in the likelihood that the insider receives good and bad news. This asymmetry may lead to a contrarian strategy of first trading against one's information and then profiting by trading in the opposite direction in the following period. Both studies thus show that mandatory disclosure leads to profitable trading opportunities for insiders and may increase expected profits, just as voluntary disclosure does in this paper. However, insiders will generally not voluntarily disclose their trades under their assumptions.

A third paper by Huddart, Hughes, and Levine (2001) obtains a contrary result by presuming that it is up to insiders to prevent others from learning their information from their trades. Informed insiders wish to do this only because they may want to trade again on the same information. While this is reasonable for mandatory disclosures, we assume that voluntary disclosures only take place once informed traders do not expect to add to their positions.

Admati and Pfleiderer (1991) consider the practice of some liquidity traders of pre-announcing the size of their trades in order to identify themselves as informationless and to help coordinate the supply and demand of liquidity in the market. They show that pre-announcement typically reduces the trading costs of those who preannounce, but its effects on the trading costs and welfare of other traders are ambiguous. Thus their model also supports the conclusion that voluntary announcements are made in equilibrium, although the purpose of such announcements is exactly the opposite of what we consider.

3 Short-Run Trading Model

The full model consists of multiple rounds of trading, where each single-round trading game is a modified Kyle (1985) model. Some of the modifications are very similar to Froot, Scharfstein, and Stein (1992). Each round considers a market for a risky asset available in fixed supply. The risky asset pays off a random amount $\tilde{v} = \tilde{a} + \tilde{b}$, where $\tilde{a}$ is a random variable equal to $+1$ or -1 with equal probability, $\tilde{b}$ is a normally distributed random variable with mean zero and variance η^2 . At the end of each complete trading round, both components of $\tilde{v}$ are made public and all positions are liquidated.

3.1 Traders and Information Structure

The main agent of our model is the *leader*, a risk neutral large trader with the ability to make announcements that influence markets, but with a significant risk of forced liquidation. A *passive investor*, also risk neutral, trades on the basis of information independent of the leader's information, but is not at risk of forced liquidation. All informed trades are masked by the presence of *noise traders* with inelastic demand for the asset. Finally, a risk neutral and competitive fringe of *market makers* fill the orders of all other traders. Market makers also keep track of the leader's reputation and respond appropriately to his announcements.⁷

All the traders submit market orders⁸ to the market makers to buy or sell the risky asset. Market makers observe only total order flow; they cannot distinguish between the orders of noise traders, the passive investor, or the leader. Because market makers are risk neutral and competitive, they earn zero profits. Key to our derivation is the assumption that *other traders are not aware ex ante of the leader's trades*. This assumption may be motivated by presuming that market makers, passive investors, and noise traders interact in a continuum of similar risky assets, but the leader only acts in one asset at a time. It is this asset that we study in each trading round, although the leader may trade in different assets at different times. The market clearing price is the market makers' expectation of $\tilde{v}$ conditional on the information contained in order flow and in the leader's public statements.

The leader may be of two possible types. He may be a *genuine leader*, in which case he receives some informative signal with frequency $f \in (0, 1)$, drawn independently in each round. He may also be a *charlatan* type, in which case he never receives a signal ($f = 0$).

⁷This is equivalent to assuming that the leader's followers, or the *public*, are themselves risk neutral and competitive. Since the orders submitted by followers are not the focus of the model, and in any case the followers would earn zero profits by the competitiveness assumption, we have folded them into the market makers.

⁸Traders submit orders before market makers determine prices.

Genuines and charlatans are identical in all respects except for their signal arrival frequency. The leader's signal $\tilde{s}$ of $\tilde{a}$ has precision $p \equiv \Pr(\tilde{a} = a | \tilde{s} = s) > \frac{1}{2}$ for $\tilde{a}$ in $\{-1, 1\}$. We consider below the situation where the leader receives a positive signal $\tilde{s} = 1$ (if any). Since this model is symmetric with respect to positive and negative signals, the reverse situation yields identical expected payoffs, and this assumption is without loss of generality.

The public believes that the leader is genuine with probability ρ , which we call his reputation. However, in the short-run model we will work with the common knowledge parameter π , the leader's credibility. Whereas reputation is defined as a property of the leader, we define credibility as a property of his messages. Credibility is defined as the ex ante unconditional probability the message $\tilde{m}$ was sent honestly. Credibility is a function of both reputation ρ and the frequency of deceptive disclosures, but we will not deal with those explicitly until later on in section 4, where we endogenize reputation in a dynamic model of repeated interaction between a given leader and the public. It is important to keep in mind that a genuine leader will be honest on some occasions and deceptive on others, so that credibility and reputation are distinct but related concepts.

The leader's early liquidation risk is parameterized by α , the probability he can hold his position until the payoff $\tilde{v}$ is announced and his private information is revealed. Thus with probability $1 - \alpha$ the leader is forced to liquidate his position early. The parameter α is a simple measure of the extent to which the leader has a high (α near zero) versus low (α near one) risk of early liquidation.

The leader would be stuck trading a very limited position, if any, were it not for his ability to announce his opinion in the form of a message $\tilde{m} \in \{-1, 1\}$ to the public.⁹ The public updates their expectation of the value of the asset on the basis of this message and the leader's reputation. If the leader is informed (i.e. receives the signal $\tilde{s} = 1$), he may send an *honest* message of $\tilde{m} = 1$. An uninformed leader (genuine or charlatan) may occasionally exploit his reputation by entering into an asset market at random¹⁰, sending a *deceptive* message and profiting from the resultant movement in price. Any leader can also choose to remain silent during any trading round.

If establishing a position is costless, then it can be shown that it is an equilibrium to continually establish random positions and send deceptive messages.¹¹ Therefore we intro-

⁹In fact, we will see later that the leader would prefer to learn $\tilde{b}$ rather than $\tilde{s}$, regardless of which is more revealing of fundamental value. This result is similar to the informational herding result of Froot, Scharfstein, and Stein (1992).

¹⁰Entry is only random from the public's point of view. The leader may have initially entered into this position deliberately, but now seeks to liquidate the position and exploits his reputation to earn some additional profits.

¹¹This equilibrium is formally defined below. We find that a credible equilibrium only exists when costs are sufficiently high.

duce a fixed per-period cost k incurred whenever the leader trades.¹² This cost may arise from either trading costs associated with establishing a position or from the costs associated with packaging and releasing a public message. Examples of the fixed cost of disclosure include time spent being interviewed in the media, production and publication costs from sending press releases or authoring articles in the financial press, and fees for money manager meetings. We assume, however, that the leader must pay the same amount regardless of whether he profits or loses, and regardless of the extent to which the market follows him. We also make the specific assumption that the cost is incurred whether or not a disclosure is made, although the results do not depend on this assumption in a significant way.

The passive investor observes information $\tilde{b}$ and trades based on this information. The investor acts independently of the leader and is not aware of his presence. Passive investing serves as the primary background informative trading and is necessary for deriving market liquidity and market depth parameters before the leader announces his presence. Without passive investors, the leader would not suffer any slippage costs and would thus take on infinite-sized positions because of risk neutrality.

As in Froot, Scharfstein, and Stein (1992), both leaders and investors have market power; thus they strategically limit their position to maximize profits. It is this strategic consideration which prevents all their information from being immediately impacted in the price. Limiting the leader's demand is also necessary in order to ensure a positive expected change in price following disclosure. We could also assume that they are risk averse and behave competitively, but we have chosen the current setup to facilitate comparison with the model of Froot, Scharfstein, and Stein. Furthermore strategic pricing power seems like a natural assumption for the leader in a model intended to capture strategic disclosures by large individual traders.

Finally, liquidity (or noise) traders submit a random demand $\tilde{u}_t$ in each period, regardless of price. We assume that the random variables $\tilde{u}_t$ are normally distributed with mean zero and variance σ^2 . Noise traders are necessary in this model to ensure a continuous price impact function. Whenever we refer to more liquid markets, we are in fact referring to a higher ratio of noise to informed trades, or $\frac{\sigma}{\eta}$.

¹²One of the ancillary effects of the fixed cost is that nondisclosure is preferable to honest disclosure when reputation is very low. For very high liquidation risk (α small), this effect is negligible for most reputation values and is a very low probability event.

3.2 Timing

Each trading round consists of a pre-announcement trading period (time 1), a post-announcement trading period (time 2), and a liquidation period. All informed traders and noise traders submit their demands based on their private information to the market makers before all trading. However, half of the passive investor’s order is executed before the announcement and the other half is executed afterwards. The assumption of staggered execution, although similar to the assumption from Froot, Scharfstein, and Stein (1992), has a different purpose in our model. We use staggered execution to derive market liquidity parameters, and it is not an essential feature of the model but rather a simplifying feature. It is the leader’s trading which is the focus, and while the leader’s trading does have an impact on prices, the market makers do not react to it directly since they do not know whether he is in the market. Nonetheless, we interpret staggered execution just as Froot, Scharfstein, and Stein do; our passive investor is a stand-in for many investors who all set out to acquire identical information,

“but receive it with a variable lag—one half of the [investors] learn the information earlier than the other half. Under this interpretation, our specification implies that [investors] must submit their demands without knowing whether they are “early” or “late” learners. We believe that similar results would emerge (although at a cost of much greater complexity) if we were to adopt the more realistic assumption that [investors] could adjust their demands *after* observing the current price and using it to make a noisy inference about the probability that they received the information relatively early or late.” [Froot, Scharfstein, and Stein (1992), p. 1464]

[Insert Figure 1 about here]

The leader is polled for his opinion once per round between time 1 and 2, at which point he make a public statemen or remain silent. If he chooses to disclose, other market participants will watch him closely and will learn whether the leader is forced into early liquidation or is permitted to hold on until his information is revealed. Since market makers are aware that the leader’s unravelling trade is uninformative, there is no price impact from this trade.

Given that an informed leader chooses to send a message, it is always in his interest to tell the truth. If he lies and the market believes him, the price would move against him and

he would end up selling at a lower average price than if he did not disclose at all. This is in contrast to Benabou and Laroque (1992), who assume that it is in the message sender's interest to convince the public of the *opposite* of what he believes. This difference arises as a direct result of differing focus as to whether the leader exploits the market through trading before or after making an announcement. The decision to manipulate markets in our model is taken before any announcement, whereas Benabou and Laroque (1992) primarily consider post-announcement trading.¹³ We extend our model to include optimal post-announcement trading in section B.1.

The competitive fringe of market makers sets prices to equal the public expectation of $\tilde{v}$, $P_t = E_t[\tilde{v}]$, where the subscript t denotes expectations taken with respect to information available at time t .

3.3 Short-Run Equilibrium

Deriving the equilibrium pricing functions and consequent demands and trading profits of the short-run model just described is a standard exercise. The reader who is interested in this derivation may now turn to Appendix A for a complete discussion of the short-run equilibrium. We make use of the following short-run equilibrium results in the dynamic model: The leader's announcement affects a direct price response of $(2p - 1)\pi$. An uninformed leader cannot profit without making a deceptive disclosure, hence he will either make no trade and no disclosure or trade and deceptive disclosure. An informed leader, on the other hand, may either make an honest disclosure or trade without disclosure. A leader who neither trades nor talks receives expected utility (trading profits) of (N stands for no trade and no disclosure)

$$U_N(\pi) = 0. \quad (1)$$

An informed leader making an honest disclosure expects to earn (H stands for honest disclosure)

$$U_H(\pi) \equiv \lambda(2p - 1)^2(\alpha + (1 - \alpha)\pi)^2 - k, \quad (2)$$

where $\lambda \equiv \left(\frac{1+\sqrt{2}}{4\sqrt[4]{2}}\right) \frac{\sigma}{\eta}$ is a market liquidity parameter directly related to $\frac{\sigma}{\eta}$. If the informed leader trades without disclosing, he expects to earn (T stands for trade without disclosure)

$$U_T(\pi) = \lambda(2p - 1)^2 \alpha^2. \quad (3)$$

¹³Benabou and Laroque (1992) relax this assumption in an extension, but the mismatch between the incentives of the message sender and receiver remain. Their paper is also the first to note the distinction between pre-announcement manipulation and post-announcement manipulation.

An uninformed leader making a deceptive disclosure expects trading profits of (D stands for deceptive disclosure)

$$U_D(\pi) \equiv \lambda(2p-1)^2(1-\alpha)^2\pi^2 - k. \quad (4)$$

Intuitively, the position entered into by the leader consists of two terms, an informational term and reputational term. An informed leader is only able to avoid early liquidation with probability α , thus the $(2p-1)\alpha$ term in (2) and (3) represents the position taken in response to the expected value of information. Any disclosing leader will gain from his immediate price reaction of $(2p-1)\pi$ only if he is forced into early liquidation (a $(1-\alpha)$ probability event), thus a $(2p-1)(1-\alpha)\pi$ term present in both (2) and (4) represents the part of the leader's position due to the expected value of reputation. Higher market liquidity λ and expected value $2p-1$ each raise the amount the leader is willing to invest and thus multiply per-unit profits, and the squared α terms come from accounting for the price impact of the leader's trades. Finally, fixed disclosure costs are subtracted to obtain the net profit equations presented above.

3.4 Endogenous Information Acquisition

In the derivation above we had assumed the leader can only learn the signal $\tilde{s}$. Consider now what happens when the leader can choose to learn either $\tilde{s}$ or $\tilde{b}$ (regarding the same asset).

Proposition 1 *Suppose the leader faces a choice of acquiring the information in either $\tilde{s}$ or $\tilde{b}$, and that the probability of early liquidation is high (α “small”). Then for any $\eta > 0$ there exists $\rho^* > 0$ such that the leader chooses to acquire $\tilde{b}$ for $\rho < \rho^*$. If α is large and η is small, the leader chooses to acquire $\tilde{s}$ for any ρ .*

This proposition states that no matter how much more significant the $\tilde{a}$ component of fundamental value relative to $\tilde{b}$, a leader with a weak reputation and very high early liquidation risk would rather learn $\tilde{b}$ over $\tilde{s}$. This is true regardless of the properties of $\tilde{a}$, $\tilde{b}$, and $\tilde{s}$. On the other hand, if the information in $\tilde{b}$ is not very valuable, then a leader who is not at great risk of forced liquidation would choose to learn $\tilde{s}$. Now that we have shown that informational herding a la Froot, Scharfstein, and Stein depends on whether the leader has a good reputation, we extend the model to a repeated interaction and analyze whether a good reputation can be sustained in the long run.

4 Dynamic Model

If reputation is a static variable, it will be in the incentive of the leader to make deceptive disclosures whenever possible. If such opportunities are plentiful, practically all of the leader's disclosures will be uninformative, so that it does not make sense for the market to attach any credibility to his messages. In contrast, allowing the public to dynamically update the leader's reputation following the outcome of each disclosure may provide the necessary incentive for leaders to voluntarily restrict their deception. Hence we can only answer whether it makes sense for the market to heed the leader's messages by embedding the short-run trading model in a multiple-round reputation game. Transitioning to a dynamic model also allows us to determine how the size of the market reaction (directly related to credibility) and price efficiency vary over time, and what is the impact of the fixed trading cost k on long-run trading profits.

[Insert Figure 2 about here]

At the start of the game, a leader is born with random type. The current probability ρ_t that the leader is of the genuine type is common knowledge; ρ_t is updated from ρ_{t-1} at the end of each trading round using Bayes' rule and ρ_1 is the public's initial prior. There are two possible ways the current leader may have sent a deceptive message. First, he may be a charlatan type, whose messages are never informative and always deceptive. Second, he may be a genuine leader who did not receive a signal in the current round but has chosen to send a deceptive message in order to exploit his reputation and earn further profits. Denote the equilibrium frequency a genuine but uninformed leader chooses to send a deceptive message with $q(\rho)$. A leader chooses $q(\rho)$ to maximize the discounted (at rate δ) expected sum of his one-round utilities. A value for $q(\rho)$ of 1 implies the genuine leader says something every time he is polled, while a value of 0 means the genuine leader speaks only when he actually has information. Values strictly between 0 and 1 denote a mixed strategy.

The charlatan type also chooses an equilibrium frequency of disclosure. However, in contrast to the genuine type, the charlatan only makes deceptive disclosures. Hence his disclosure frequency $\phi(\rho)$ is comparable to the genuine type's overall disclosure frequency (honest and deceptive) of $f + (1 - f)q(\rho)$. If the charlatan sends messages with a different overall frequency from the genuine leader, then the fact that a disclosure either did or did not take place is itself informative of the leader's type. Hence reputation ρ is immediately

updated to

$$\rho_D(\rho) \equiv \frac{(f + (1 - f)q)\rho}{(f + (1 - f)q)\rho + \phi(1 - \rho)} \quad (5)$$

upon observing disclosure and to

$$\rho_N(\rho) \equiv \frac{(1 - f)(1 - q)\rho}{(1 - f)(1 - q)\rho + (1 - \phi)(1 - \rho)} \quad (6)$$

if the leader remains silent. Note that no updating takes place if the leader is known for sure to be a charlatan ($\rho = 0$) or genuine ($\rho = 1$). For any initial reputation in between, disclosure may increase or decrease reputation, depending on whether ϕ is less or more than $f + (1 - f)q$, respectively. If $\phi = 1$, implying the charlatan always gives his opinion whenever polled, then a single instance of nondisclosure would lead the public to immediately conclude that the leader is genuine ($\rho_N = 1$). At the other extreme if $\phi = 0$, a single disclosures proves that the leader is genuine ($\rho_D = 1$). Later on, we will introduce conditions such that this does not occur in equilibrium. In fact, we will provide conditions such that $\phi = f + (1 - f)q$, i.e. that the charlatan mimics the genuine leader, whenever $q > 0$.

The passive investor and noise traders behave as successive generations of identical, one-round traders. The leader's reputation ρ_t is the only intertemporal link between trading rounds. At the end of round t , the true $\tilde{a}$ and $\tilde{b}$ are observed and the leader's reputation is updated. The leader expects to receive a trading profit of $U_H(\pi)$ if he sent an honest message, $U_D(\rho)$ if he sent a deceptive message, and $U_N(\rho) = 0$ if he was silent.

If the reputation at date t is ρ_t , then a genuine leader will send some message with frequency $f + (1 - f)q(\rho_t)$. Of these messages, the disclosure is honest f of the time and deceptive $(1 - f)q(\rho_t)$ of the time. Honest predictions are correct with probability p and deceptive predictions are correct with probability $\frac{1}{2}$. Thus the probability a genuine leader's prediction is correct is

$$\frac{fp + \frac{1}{2}(1 - f)q(\rho_t)}{f + (1 - f)q(\rho_t)}. \quad (7)$$

On the other hand, a charlatan's predictions are correct only half the time. The unconditional probability a prediction is correct is the sum of these two conditional probabilities weighted by $\rho_D(\rho_t)$, the updated reputation given disclosure, or

$$c(\rho_t) \equiv \frac{\rho_D(\rho_t)f(p - \frac{1}{2}) + \frac{1}{2}(f + (1 - f)q(\rho_t))}{f + (1 - f)q(\rho_t)}. \quad (8)$$

Recall from section 3.1 the notation π for the credibility, or the unconditional probability

a given disclosure is honest. Dividing the unconditional frequency of honest disclosures by total disclosures, we find the following relation between credibility π , reputation ρ , and deception frequency q :

$$\pi(\rho_t) \equiv \frac{\rho_D(\rho_t) f}{f + (1-f)q(\rho_t)}. \quad (9)$$

If the genuine type always receives a signal ($f = 1$), credibility is equivalent to reputation following disclosure. In general, credibility will be lower for lower signal reception frequencies f than for higher f . The probability of being correct (8) may alternately be derived as the sum $\pi(\rho_t)p + (1 - \pi(\rho_t))\frac{1}{2}$. Applying Bayes' rule and the definition of reputation, we find the public further updates reputation following the realization of information $\tilde{a}_t$ as

$$\rho_{t+1} = \begin{cases} \rho^+(\rho_t) \equiv \left(1 - \frac{(1-\rho_t)f(2p-1)}{2fp+(1-f)q(\rho_t)}\right)^{-1} \rho_D(\rho_t) & \text{after a correct forecast } (a_t = m_t) \\ \rho^-(\rho_t) \equiv \left(1 + \frac{(1-\rho_t)f(2p-1)}{2f(1-p)+(1-f)q(\rho_t)}\right)^{-1} \rho_D(\rho_t) & \text{after an incorrect forecast } (a_t = -m_t) \end{cases} \quad (10)$$

If no disclosure was made, reputation is not updated based on $\tilde{a}_t$, and so the next period's reputation ρ_{t+1} is $\rho_N(\rho_t)$. Note from the expressions above that reputation strictly increases with each correct prediction and decreases with each incorrect one, unless the leader is known for sure to be genuine ($\rho_t = 1$) or a charlatan ($\rho_t = 0$). In contrast to the standard reputation literature (see Sobel (1985)), even when the signal is perfect ($p = 1$), reputation does not immediately collapse to $\rho_t = 0$ following an incorrect prediction. Rather, only if either the genuine type chooses never to disclose deceptively ($q(\rho_t) = 0$) or he always receives a signal ($f = 0$) will an incorrect prediction constitute conclusive evidence that the current leader is a charlatan. In such a case, the charlatan's reputation improves monotonically until the first incorrect prediction, after which $\rho_t = 0$ forever. When either the signal is noisy ($p < 1$) or the genuine leader makes deceptive disclosures, reputation evolves more realistically, with both ups and downs but never reaching either 0 or 1. Similarly to other papers, reputation ρ_t embodies all information available to the public at the start of round t (denoted Ω_t), hence it follows, from the public's point of view, a martingale:

$$E[\rho_{t+1}|\Omega_t] = c(\rho_t)\rho^+(\rho_t) + (1 - c(\rho_t))\rho^-(\rho_t) = \rho_t. \quad (11)$$

The objective process of ρ_t along an equilibrium path depends on the leader's type; it will be characterized in section 5.

4.1 Reputation Equilibrium

The broader logic below proceeds somewhat similarly to Benabou and Laroque (1992).¹⁴ The genuine leader must now take into account the effect of making disclosures on both his future reputation as well as his immediate profit. Suppose that at time t he does not receive the signal. If he makes a deceptive disclosure, he will receive immediate utility $U_D(\pi(\rho_t))$. With probability $\frac{1}{2}$ his prediction will be correct and his reputation in the next period will increase to $\rho_{t+1}^+(\rho_t)$, while with probability $\frac{1}{2}$ it will be wrong, and his reputation will decrease to $\rho_{t+1}^-(\rho_t)$. Alternatively, an uninformed leader may choose not to trade and not to make any public statements, in which case he receives no immediate utility and his reputation is updated to $\rho_N(\rho_t)$.

Suppose, on the other hand, that the leader receives a signal in the current round. Then he may send a message containing an honest disclosure of his signal, in which case he receives immediate utility $U_H(\pi(\rho_t))$ and his reputation increases to $\rho_{t+1}^+(\rho_t)$ when he is correct (with probability p) and decreases to $\rho_{t+1}^-(\rho_t)$ otherwise. However, if immediate profits from trading without disclosure U_T are higher than U_H by more than the amount of reputational benefit from honest disclosure (e.g. from very high disclosure costs k), then he may trade on the signal without disclosing it. In such a case, he receives immediate utility U_T and his reputation in the following round is $\rho_N(\rho_t)$. We show in Lemma 2 that an informed leader prefers to trade without disclosure as ρ approaches zero.

Let $W(\rho_t)$ be the discounted expected sum of the genuine type's utility at the beginning of a round before he knows whether or not he receives a signal (this function implicitly depends on $q(\rho_t)$ and $\phi(\rho_t)$). When the genuine type receives a signal, he may disclose (honestly) and get

$$V_H(\rho_t) = U_H(\pi(\rho_t)) + \delta(pW[\rho^+(\rho_t)] + (1-p)W[\rho^-(\rho_t)]) \quad (12)$$

or he may remain silent but trade

$$V_T(\rho) = U_T(\pi(\rho)) + \delta W(\rho_N(\rho)) \quad (13)$$

When he does not receive the signal, if he takes a position at random and sends a message he gets

$$V_D(\rho_t) = U_D(\pi(\rho_t)) + \frac{\delta}{2}(W[\rho^+(\rho_t)] + W[\rho^-(\rho_t)]) \quad (14)$$

¹⁴Certain portions below are paraphrased from their paper's section IV.B, making the appropriate substitutions for this model.

whereas his payoff if he remains silent and stays out of the market is

$$V_N(\rho_t) = \delta W(\rho_N(\rho_t)) \quad (15)$$

[Insert Figure 3 about here]

The uninformed leader weighs his immediate profit-related *incentive to deceive* $U_D(\pi(\rho_t))$ against his reputational *incentive to be silent* $\delta W(\rho_t) - \frac{\delta}{2}(W(\rho_{t+1}^+) + W(\rho_{t+1}^-))$. If $V_D(\rho_t) > V_N(\rho_t)$, he always deceives, $q(\rho_t) = 1$ and $\phi(\rho_t) = 1$, and the credibility of messages is $\pi(\rho_t) = \rho_t f$. When $V_D(\rho_t) = V_N(\rho_t)$, he randomizes between deceptive disclosure and nondisclosure such that the genuine type's deceptive disclosure frequency $q(\rho_t) \in [0, 1]$ and the charlatan type's disclosure frequency $\phi(\rho_t) \in [0, 1]$. If $V_D(\rho_t) < V_N(\rho_t)$, a leader never makes deceptive disclosures. However, as we will show below, this can never take place in equilibrium for then all disclosures are honest, and a single disclosure constitutes proof of type. If the public updated reputation following any disclosure in this way, the value of deception would certainly be higher than the value of nondisclosure, a contradiction. Before formally proving this statement, we must define equilibrium with the following proposition:

Proposition 2 *An equilibrium of the dynamic reputation game corresponds to a disclosure strategy of the genuine and charlatan types as functions $q, \phi : [0, 1] \rightarrow [0, 1]$ of reputation ρ and an associated value function $W : [0, 1] \rightarrow \mathbb{R}$ such that for all ρ :*

$$\begin{cases} V_D(\rho) < V_N(\rho) & \text{implies that } q(\rho) = \phi(\rho) = 0, \\ V_D(\rho) = V_N(\rho) & \text{implies that } 0 \leq q(\rho) \leq 1 \text{ and } 0 \leq \phi(\rho) \leq 1, \\ V_D(\rho) > V_N(\rho) & \text{implies that } q(\rho) = \phi(\rho) = 1, \end{cases} \quad (16)$$

$$W(\rho) = f \max\{V_H(\rho), V_T(\rho)\} + (1 - f) \max\{V_D(\rho), V_N(\rho)\} \quad (17)$$

where the functions V_H , V_T , V_D and V_N are defined from W by (12)-(15), from q by (9) and (1)-(4), and from ϕ by (5) and (6).

We concentrate on equilibria associated with a value function $W(\rho)$ of the genuine leader which is continuous and nondecreasing. C_+ will denote the space of such functions on $[0, 1]$, endowed with the norm of uniform convergence.

We now proceed to solve for the equilibrium functions W , q , and ϕ . Consider what happens for a given ρ and W . Suppose the genuine leader chooses some deception frequency q . Then charlatan will choose ϕ such that $V_D(\phi) = V_N(\phi)$. It is clear that ρ_D is decreasing

in ϕ , whereas ρ_N is increasing. By monotonicity of W , this implies V_D is decreasing and V_N is increasing in ϕ . Hence there is a unique $\phi(q)$ defined by $V_D(\phi) = V_N(\phi)$ whenever such a point exists for $\phi \in (0, 1)$ and by $\phi = 0$ or 1 for $V_D(0) < V_N(0)$ and $V_D(1) > V_N(1)$, respectively.

Proposition 3 *The disclosure frequency of the charlatan type, ϕ , equals the genuine type's total disclosure frequency, $f + (1 - f)q$, for a sufficiently high discount factor and reputation. The public does not infer any new information regarding type from lack of disclosure whenever disclosure frequencies are equal.*

Intuitively, if disclosure frequencies differed, the public would arrive at a more accurate estimate of the leader's type merely by comparing the realized frequency of disclosure to the predicted frequency. If disclosure opportunities are spaced very close together, it is in the charlatan's interest to mimic the genuine type's disclosure frequency in order to stave off detection.

Henceforth, we will take δ very close to one such that the charlatan sends messages with the same overall frequency as the genuine type, and so nondisclosures are not informative events. Taking the discount factor to be very close to one may also be interpreted as having very short trading rounds or very frequent opinion polling. At the same time we will take f very close to zero, so that the genuine leader faces many uninformed decisions for each informed decision. These assumptions essentially transforms our discrete time model into something like a continuous time setting with a positive information arrival rate $\frac{f}{1-\delta}$. Such a continuous time setting improves the realism of the model, as real-world leaders can effectively choose to disclose their opinions at any time, but have limited information gathering abilities.

We also make use of the above proposition to set $\phi = f + (1 - f)q$ in equilibrium for all ρ such that $q(\rho) > 0$. Whenever $q(\rho) = 0$, we set ϕ to some value between zero and f such that $V_D = V_N$. Since our focus is on the behavior of the genuine type whenever he chooses to make some deceptive disclosures, we do not investigate the properties of $\phi(\rho)$ over the full domain.

In the short run, where the reputational value W from future trading is taken as given, we find that each reputation ρ is associated with a unique equilibrium disclosure frequency q :

Proposition 4 *There exists $k^* > 0$ such that for $k > k^*$ the short-term game has a unique equilibrium; the genuine type's optimal deception frequency is nondecreasing in reputation. Formally, for any ρ in $[0, 1]$ and W in C_+ , there exists a unique $q^*(\rho, W)$ satisfying 16; q^* is continuous in both arguments and nondecreasing in ρ . The functions $V_H^*(\rho, W)$, $V_T^*(\rho, W)$, $V_D^*(\rho, W)$, and $V_N^*(\rho, W)$ on $[0, 1] \times C_+$ associated with W and q^* by (12)-(15) are continuous and nondecreasing in both arguments.*

[Insert Figure 4 about here]

The proposition above tells us that a well-reputed leader has a greater incentive to exploit his reputation by making deceptive disclosures. Intuitively, a better reputation raises the value of deceptive disclosure, V_D , more than the value of nondisclosure, V_N . Holding long-run reputational value W fixed, raising the frequency of deceptive disclosures lowers V_D but not V_N . If the leader is to mix between disclosure and nondisclosure ($q \in (0, 1)$), he must raise the equilibrium disclosure frequency in order to maintain indifference.

The result requires that fixed disclosure costs k be higher than some k^* in order to ensure the uniqueness of the equilibrium. Disclosure costs in this model act as a commitment mechanism for the leader. The leader's profitability depends almost entirely on his credibility. However the loss in credibility from each additional disclosure is outweighed by the additional profit. Thus if disclosure costs are very low or zero, the equilibrium will unravel and the leader will choose $q = 1$, or maximum deceptive disclosure. Sufficiently high disclosure costs are needed to guarantee a turning point in disclosure policy as a function of profits.

To solve for the long-run equilibrium, we impose the restriction that the function W indeed be the expected present value of future trading profits. Let T be the mapping that associates to any end-of-period valuation W the beginning-of-period valuation resulting from the short-term game, i.e.,

$$T(W) : \rho \in [0, 1] \rightarrow f \max \{V_H^*(\rho), V_T^*(\rho)\} + (1 - f) \max \{V_D^*(\rho, W), V_N^*(\rho, W)\}. \quad (18)$$

The continuity and monotonicity properties of Proposition 4 allow us to use Blackwell's theorem and show that T is a contraction on C_+ .

Proposition 5 *There is a unique Markovian equilibrium of the dynamic game, with a continuous, nondecreasing value function W .*

[Insert Figure 5 about here]

A rising reputation exerts two opposing forces on the equilibrium credibility. The greater likelihood the current leader is genuine, represented by the term ρf in (9), raises equilibrium credibility. On the other hand, the higher reputation results in more frequent deception, represented by the term $(1 - f) q(\rho)$. Determining the net impact on credibility is very important for it is directly related to the size of the price reaction to the leader's messages (see (27)).

4.2 Numerical Simulation

Further properties of the model are explored using numerical simulation. Here we describe the process of numerical approximation and convergence used to attain the graphs presented throughout the paper. Some of the propositions below are also based on numerical simulation. A full analytical proof of these results is left for future work.

We explore the solution to the model for the following parameter values: $f = 0.1$, $\lambda = 10$, $\alpha = 0.1$, $\delta = 0.99$, $k = 0.1$, and $p = 0.75$. By the properties of $W(\rho)$ proven in Proposition 5, W can be attained as the limit of successive approximations and reoptimizations. The optimal $q(\rho)$ is set so that (16) is satisfied. Apply Lemmas 2 and 1 to find function values at the endpoints of the domain of ρ . At the lower endpoint, we find $\lim_{\rho \rightarrow 0} q(\rho) = 0$, and so $\lim_{\rho \rightarrow 0} \pi(\rho) = 0$. We have taken $k \geq \lambda \alpha^2$, so that by (17) we find $\lim_{\rho \rightarrow 0} W(\rho) = 0$. Proceed by dividing the unit interval of possible ρ into some number N of grid points and positing a linear $W(\rho)$ between the two endpoints. Each iteration then derives $q(\rho)$ from (16) on the basis of the $W(\rho)$ from the previous iteration, then calculates the next iteration of $W(\rho)$ from the newly optimized $q(\rho)$. We calculate a new $W(\rho)$ from $q(\rho)$ and the current $W(\rho)$ as many times as is necessary to converge to some $W(\rho)$ to within 1×10^{-7} .

Our numerical calculations show that a k greater than some unknown positive quantity is needed to guarantee the unique equilibrium described in Proposition 5. It also appears that the necessary lower bound on k approaches 0 as $p \rightarrow \frac{1}{2}$.

5 Transmission of Information in the Long Run

Earlier we demonstrated that the reputation process $\{\rho_t\}$ is a martingale with respect to $\{\Omega_t\}$, the public's information set. But when the leader is in fact genuine (denoted by G),

the true transition probability is

$$E [\rho_{t+1} | \Omega_t, G] = p\rho^+ (\rho_t) + (1 - p) \rho^- (\rho_t) \geq \rho_t$$

with strict inequality when ρ_t is different from 0 or 1, and where the inequality follows from (11) and the fact that $c(\rho_t) \leq p$. In other words, a genuine leader's reputation tends to improve and is in fact a strict submartingale. As we will show below, for any initial condition ρ_1 , it converges almost sure to 1. Similarly, as the charlatan's predictions are correct only half the time, less than $c(\rho_t)$ for $\rho_t \in (0, 1)$, his reputation will deteriorate on average. When the leader is a charlatan (denoted by C), ρ_t is really a supermartingale:

$$E [\rho_{t+1} | \Omega_t, C] = \frac{1}{2}\rho^+ (\rho_t) + \frac{1}{2}\rho^- (\rho_t) \leq \rho_t \quad (19)$$

with strict inequality for $\rho_t \in (0, 1)$. The following proposition proves these results rigorously.

Proposition 6 *The leader reveals his type asymptotically: from any initial reputation ρ_1 in $(0, 1)$, the equilibrium reputation process $\{\rho_t\}$ converges almost surely to 0 (respectively, to 1) as t goes to infinity if the leader is a charlatan (respectively, genuine). Convergence becomes faster for genuine types in liquid markets ($\lambda > \frac{k}{4}$).*

Hence we can define the long-run discounted trading profit of the genuine leader $\bar{W}$ as the limit taking $f \rightarrow 0$ and $\delta \rightarrow 1$ holding $\frac{f}{1-\delta}$ constant of $\lim_{\rho \rightarrow 1} W(\rho)$.

Proposition 7 *The long-run discounted trading profit of the genuine leader $\bar{W}$ is increasing in fixed trading cost k for $k < \lambda(1 - \alpha)^2$. It is also increasing in market liquidity λ , non-liquidation probability α , signal precision p , and signal frequency $\frac{f}{1-\delta}$.*

Of the results proven above, none are very surprising except for the fact that profits are increasing in k , a cost borne by the leader every time he makes a trade. Although a higher trading cost lowers the profit for each trade and disclosure event, it also inhibits deception. In equilibrium, a lower rate of deception increases the credibility of each announcement and the size of the market reaction. The proposition above shows that the net effect of these two forces results in an increase in discounted trading profits.

Another interesting result concerns the evolution of price efficiency over time. Define the average efficiency of prices as the negative of the squared difference between post-announcement price and fundamental value incorporating public and privately known infor-

mation, averaged by frequency across all types of periods (including those with no disclosure), or

$$\mathcal{E}(\rho) \equiv - \left(f(1 - \pi(\rho))^2 + (1 - f)q(\rho)\pi(\rho)^2 \right) \quad (20)$$

Proposition 8 *Price efficiency is decreasing in ρ (i.e. $\mathcal{E}'(\rho) < 0$) for ρ values such that deception is possible ($q(\rho) > 0$).*

This result shows that better knowledge regarding a leader's type may be harmful to markets subject to the type of manipulation we have examined here. Combined with Proposition 6, which shows that reputation is generally increasing for genuine leaders, this tells us that as the public gathers more information concerning a genuine leader, prices actually become less efficient.

6 Empirical Findings

In this section we describe a framework for testing the empirical implications of the model. Our goals for empirical testing are twofold. First we seek to confirm the presence of stock price reactions to voluntary disclosure, which provides additional motivation for the underlying mechanism behind all our results. Second, we adapt some of the propositions above to the empirical framework and test the model's predictions.

6.1 Data and Methodology

Most U.S. mutual fund managers (and many foreign managers) disclose all of their holdings (at least) quarterly in a mandatory report to the SEC. However, they may also opt to speak publicly about their latest acquisitions before they are disclosed. We would expect the results from the model above to apply less strongly, if at all, to those funds with relatively fewer mentions in the financial press. Looking back on past mandatory disclosures, we can also tell retrospectively which stocks a fund was buying or selling, and whether managers were establishing new positions or adding to existing positions. We can even tell whether such trades took place over a relatively long or short period. Combining information from mandatory SEC disclosures with information on voluntary disclosure in the financial press allows us to build a database from which many of the model's assumptions and implications can be tested.

Our database contains quarterly information on mutual funds' stock holdings and returns on those stocks over the eleven-year period 1995-2005. We collect information on the U.S.

equity holdings of all registered mutual funds filing with the SEC plus a large number of global funds from Thomson Financial’s CDA/Spectrum Mutual Funds Holding database. Each observation consists of the number of shares held as of a given “report date,” as well as the net change since the prior report date.¹⁵ Most funds report every quarter, although some occasionally report at a different frequency. We drop all funds with more than one quarter (93 days) since the last report date (20% of the original sample). Of the remaining observations, roughly 10% of funds last reported one month ago, 5% reported two months ago, and the remainder reported a full quarter ago. Thomson also reports when the information contained in each report became public, known as the file date.¹⁶

We drop all funds with total assets under management of under \$50 million, as well as funds classified as “international” funds, “bond” funds, or unclassified. We only keep U.S. common stock holdings which are traded on the NYSE, AMEX, or NASDAQ and which are not closed-end funds, Americus Trust Components, or REITs. Furthermore, we drop stocks which constitute less than 2% of the fund’s assets, for the portfolio manager is unlikely to spend much effort spreading the word on holdings which do not contribute significantly to his returns. Since our goal is to analyze returns for these stocks, we also drop illiquid stocks, which we define as those with peak market capitalization below \$100 million, a peak price below \$5, and average daily volume less than 100,000 shares. This leaves us with a database of 1,735 funds and 4,234 stocks. Each observation contains information for one stock held by a particular fund (i.e. if ten funds hold the same stock on a given date, ten different observations will have information for that stock on that date), and there are 224,630 observations in total.

We combine these data with information for each stock on daily holding period returns, volume, shares outstanding, and SIC code from the Center for Research in Securities Prices (CRSP). Using these data, we calculate cumulative abnormal returns (CARs) around the report dates as follows. Each stock is assigned to a Fama-French (50) industry category based on SIC code. We use each individual year’s sample to calculate the loadings of these fifty portfolios using the Fama and French (1993) three-factor model (factors obtained from Kenneth French’s website). The loadings from one year are used to calculate daily abnormal returns (net of the risk-free rate) for the following year. Abnormal returns are then cumulated over the period from the prior report date to the current report date to form

¹⁵Unfortunately Thomson does not keep track of the prior report date until 1999, and so variables dependent on the exact prior report date begin only in 1999.

¹⁶Roughly one in three mutual funds files with the SEC more than a few days after the report date.

the before-report CAR (BCAR) variable.¹⁷ We also calculate a forward-looking CAR using days 0 to +20, +50, +100, and +200 relative to the report date (FCAR20, FCAR50, ...). All CARs are winsorized at the first and ninety-ninth percentile levels.

The final and most important data are the numbers of financial press articles mentioning a particular mutual fund manager collected by searching Lexis-Nexis. For each mutual fund, we searched for the terms “fund,” “manager,” and the long form name of the fund according to CDA/Spectrum. In most cases the long form name of the fund is obvious, although some funds, such as “RS INV MGMT EMERGING GW,” presented more of a challenge and some discretion had to be applied. Certain fund names which made for very common search terms, such as “The Fidelity Fund,” were also excluded.

The total number of LexisNexis hits in the “Business & Finance” news source database from the eleven year period examined are recorded as the variable LEXISFUND. We use this variable as a proxy for the degree to which the portfolio manager of the fund makes voluntary disclosures regarding his positions to the press. Although we recognize that there are many potential reasons a fund’s name may appear in an article. Since portfolio managers frequently manage many mutual funds across a fund family (e.g. Fidelity, Putnam), and they may make strategic disclosures to benefit all of their funds, we construct LEXISFAM as the average of LEXISFUND for all funds in a given fund family. Although work on collecting these data is ongoing, we have thus far collected information for 400 funds, emphasizing the largest funds and funds with high turnover. Data for these funds represents roughly 25% of mutual fund holdings observations.

6.2 Evidence for Stock Price Reactions to Voluntary Disclosure

The most fundamental and essential mechanism of the model is the immediate stock price reaction to a voluntary disclosure. We test for the presence of this reaction amongst stocks that some portfolio manager has recently acquired. Specifically, we test the following hypothesis

Claim 1 *Stocks purchased by portfolio managers who frequently disclose their holdings to the financial press receive a larger immediate price reaction than those stocks purchased by silent fund managers.*

¹⁷Due to computer processing limitations, we use the following easy-to-construct approximation: For observations with fewer than 32 ordinary days between reports we use a 22 trading day cumulation, whereas for observations with 59 to 93 ordinary days we use a 64 trading day cumulation. No observations have between 32 and 58 days between reports.

The key variable for testing this hypothesis is *NEWBUY*, an indicator for whether the stock did not appear in the particular fund’s prior report. Recall that each fund in our dataset has one observation for each stock (provided it constitutes at least 2% of total assets) and for each disclosure period. Thus if one fund buys a stock for which another fund already owns a significant share, the same stock will have differing values of *NEWBUY*.

The central test of the claim above is a regression of *BCAR* on the interaction of *NEWBUY* and *LEXISFUND*. Since it may be expected that the fund’s own buying will influence the stock price, we also restrict the sample for these regressions to observations for which the fund’s average daily buying constituted less than 0.5% of average daily volume. If voluntary disclosure impacts market prices, the highest immediate reactions to new buys should come to those stocks bought by mutual funds with a relatively high *LEXISFUND*. Thus we test the voluntary disclosure mechanism by estimating the following specification

$$\begin{aligned} BCAR_{i,t} = & \alpha + \beta_1 NEWBUY_{i,t} + (LEXISFUND2_i + LEXISFUND3_i) \\ & + (\beta_2 LEXISFUND2_i + \beta_3 LEXISFUND3_i) * NEWBUY_{i,t} \quad (21) \\ & + QUARTERdummies_t + FUNDSIZEdummies_i + \varepsilon_{i,t} \end{aligned}$$

The left-hand side (LHS) variable is *BCAR*, the cumulative abnormal return before the report date. The coefficients of interest are $\beta_1 - \beta_3$, measuring the stock price reaction due to *NEWBUY* across terciles (three quantiles) of *LEXISFUND* (*LEXISFUND1* is dropped from the regression). *QUARTER* and *FUNDSIZE* represent the current quarter and the fund size quintile by assets under management. Dummies of each of these are included as right-hand side (RHS) control variables. The results are reported in Table 1. Column 1 estimates the above specification for the entire sample and column 2 contains estimates for a subsample of stocks with market capitalization less than \$20 billion. Since disclosure is expected to have a larger impact on smaller stocks, we expect the latter column to contain more significant estimates. The particular cutoff of \$20 billion is arbitrary and similar results are obtained for other cutoff values. Columns 3 and 4 estimate the same specifications as columns 1 and 2 using the Fama-MacBeth approach instead of quarter dummies.

The estimates presented in Table 1 show that β_3 is positive, as expected. It is also both highly statistically and economically significant in all four specifications, leading us to believe the disclosure effect applies mostly to acquisitions of small or mid capitalization stocks. The estimate of β_3 in column 4 suggests an immediate price reaction of as much as

13% from voluntary disclosure. On the other hand, the estimates of β_1 and β_2 are much lower in magnitude and statistical significance, suggesting the fund's own buying effect on the stock is comparatively small.

We are worried that the primary effect captured by LEXISFUND is not so much a voluntary disclosure effect as it is a fund size effect. Indeed, a regression of LEXISFUND on assets under management yields an R^2 of 31% and a very significant coefficient (t-stat of 213). We therefore construct a purer proxy for voluntary disclosure, LEXRESID, from the residual of this regression. Table 2 presents results from estimating the same specification as (21) with terciles of LEXRESID substituted for LEXISFUND. The coefficient estimates clearly show that most of the LEXISFUND effect was not due to variations in fund size. We are similarly worried that LEXISFUND tends to pick up on past performance, although a regression of LEXISFUND on various measures of past performance did not yield substantive values for R^2 .

Another potential concern with either specification above is selection bias. Namely, the financial press is more likely to write articles about funds that do well. Hence funds may be in the news because of good recent returns (i.e. BCAR), and not so much because their holdings do well as a result of voluntary statements by the portfolio manager. However, we believe this problem is mitigated by collecting the number of hits over the entire eleven year period rather than just around the time BCAR was measured. Furthermore, most articles mentioning the portfolio manager tend to discuss both old and recently acquired positions, and it is exceedingly rare that an article is written on the basis of a single recent acquisition.

Even if we believe the number of Lexis-Nexis hits is driven by the performance of the fund's holdings, the contemporaneous effect of returns on LEXISFUND should be picked up by the constants in the regression, not the interaction terms we examined.¹⁸ Indeed most of these constants are positive and significantly higher for LEXISFUND3 than LEXISFUND2. Potential problems are further mitigated by analyzing LEXISFAM in place of LEXISFUND. Regressions similar to those above estimated using LEXISFAM yield similar results, but of lesser magnitude and statistical significance. Finally, we can avoid this problem altogether by collecting year-by-year or quarter-by-quarter Lexis-Nexis search results and using only hits prior to the current quarter. This is left for future work.

¹⁸Remember that the other observations in the sample for which NEWBUY is zero are also 2% holdings of the fund.

6.3 Testing Further Implications

Our dataset also allows us to test for mandatory disclosure effects by looking at abnormal stock returns following the public disclosure of a fund’s holdings to the SEC. Our model predicts that the stock price reaction to disclosure is stronger for leaders with powerful reputations. Hence we construct a proxy for reputation using past performance. The indicator variable *PERF* is equal to one for funds in the top five percentile by increase in assets under management from two quarters ago to the previous quarter and zero for those in the bottom ninety-five percentile. This group is select enough that we expect many market participants to follow the leader yet large enough to yield statistically significant estimates.

We estimate the following specification to test for reputation effects on mandatory disclosure

$$\begin{aligned} FCAR_{i,t} = & \alpha + \beta_1 NEWBUY_{i,t} + \beta_2 PERF_{i,t} + \beta_3 NEWBUY_{i,t} * PERF_{i,t} \\ & + QUARTERdummies_t + FUNDSIZEdummies_i + \varepsilon_{i,t} \end{aligned}$$

The LHS variable is *FCAR*, the cumulative abnormal return following the mandatory disclosure date. We estimated the specification for a few different horizons beyond the report date, and significance was greatest for a horizon of 50 (trading) days past report date. The RHS variable of interest is the interaction term between *NEWBUY*, the new buy indicator, and *PERF*, the past performance indicator. We expect the coefficient on the interaction to be positive if stock prices react more to the recent purchases of managers that have performed well in the recent past. While the interaction term was consistently positive across different subsamples restricted by market capitalization, significant estimates were only obtained when dropping stocks with market capitalization below \$700 million and funds in the lowest size quintile by assets under management. These estimates are presented in Table 3. Column 1 estimates the specification using the pooled sample while column 2 uses the Fama-MacBeth approach and drops quarter dummies. While these results obtained only for a specific subsample are admittedly weaker, we believe it at least provides some motivation for further tests of the effects of reputation.

Empirical testing and calibration of the model is by no means complete. We intend to use this dataset to test some of the novel predictions presented in section 5, such as the effect of reputation on accuracy of predictions and on price efficiency. In particular, we expect that better reputation leads to more reversals, and that the size of the reversal should be

proportional to the initial rally. We may also test more directly the prediction that more liquid stocks experience smaller disclosure effects by introducing one of the popular proxies for liquidity and controlling for market capitalization.

Another testable prediction is that reputation mitigates herding amongst mutual fund managers. If managers with good reputations make use of those reputations to pursue more obscure information, such managers should also have a greater variety of stocks in their portfolio and a greater number of less commonly held stocks. However, this prediction can also be the result of mutual fund managers' career concerns, as suggested by Chevalier and Ellison (1999). Their paper also provides evidence for this prediction. On the other hand, Prendergast and Stole (1996) suggest that younger managers may exaggerate their use of private signals in an effort to appear as fast learners. Relating the data on herding in mutual fund holdings to our voluntary disclosure data may help resolve the debate over how portfolio managers behave and why.

The tests of the model described above take a macro approach, screening for either voluntary disclosure frequency or reputation. However, the model is also amenable to micro testing. By collecting an exhaustive list of all of a particular trader's voluntary disclosures, one can test the leader-specific predictions such as how disclosure frequency and prediction accuracy track reputation. One of our intended avenues for future work is to collect such data by searching LexisNexis for the names of famous money managers such as Peter Lynch or George Soros and tracking what they say (positive or negative) in every article. In the case of Peter Lynch, one can even track how his positions affected what he said. We hope that such work can lead to valuable lessons on evaluating future statements by portfolio managers.

7 Conclusion

We develop a dynamic model of a repeated interaction between a privately informed trader and the public. The trader wishes to minimize the risk that an adverse price move forces him to liquidate before his information is realized by disclosing the information and profiting immediately from the market price reaction. The public (i.e. the market) attempts to discern whether the trader is a genuine leader with access to valuable information signals, or a charlatan who merely pretends to have information. A genuine leader will exploit his reputation some of the time by disclosing trades he intends to liquidate that are not

based on private information. The leader balances the additional profits from deceptive disclosure against the probabilistic loss in reputation. Since the leader optimally restricts the frequency of exploitation, the public will heed the leader’s advice. Hence our main result is that voluntary trade disclosure may mitigate early liquidation risk and restore the incentive to gather private information.

Our model yields a number of additional empirical implications regarding the credibility of voluntary disclosures. Although leaders with better past performance records are more likely to possess an informational edge, they are also more likely to exploit their reputation by making more deceptive disclosures. Hence further improvements in credibility are at best marginal for those leaders who already have excellent reputations. Disclosure in a more costly forum leads to both a higher prediction accuracy and increased profits for the leader. Increasing the opportunities for disclosure need not result in a commensurate increase in exploitative behavior, particularly if opportunities were already plentiful. The market reaction is lower for more liquid assets. Leaders with better reputations keep their best information private. Finally, the market reaction to genuine leaders may become less efficient over time as leaders with better reputations manipulate markets more often.

Since credibility is directly related to the market price reaction, many of the results just outlined are amenable to statistical testing. To this end, we gather a panel dataset on mutual fund holdings, returns on those holdings, and frequency of financial press articles mentioning the portfolio manager, a proxy for voluntary disclosure. Consistent with this model, we find that, controlling for certain characteristics of the mutual funds’ holdings, there is a significant stock market reaction to voluntary disclosure. We also find that the disclosure effect is stronger for funds with better recent past performance, although long-term performance appears less relevant.

There are a number of directions available for future research on the informational content of voluntary disclosures. Our dataset can be expanded and utilized to test many of the model’s other implications mentioned above. We can also collect manager-specific data to test the model’s results on the time-path of deceptive and honest disclosure. We plan to pursue these avenues in future research.

References

- Abreu, Dilip, and Markus Brunnermeier, 2002, Synchronization risk and delayed arbitrage, *Journal of Financial Economics* 66, 341–360.
- Admati, Anat, and Paul Pfleiderer, 1991, Sunshine trading and financial market equilibrium, *Review of Financial Studies* 4, 443–481.
- Benabou, Roland, and Guy Laroque, 1992, Using privileged information to manipulate markets: Insiders, gurus, and credibility, *Quarterly Journal of Economics* 107, 921–958.
- Biggs, Barton, 2006, *Hedgehogging* (John Wiley and Sons: Hoboken, New Jersey).
- Brunnermeier, Markus, and Lasse Heje Pederson, 2005, Predatory trading, *Journal of Finance* 60, 1825–1863.
- Chevalier, Judith, and Glenn Ellison, 1999, Career concerns of mutual fund managers, *Quarterly Journal of Economics* 114, 389–432.
- Corsetti, Giancarlo, Amil Dasgupta, Stephen Morris, and Hyun Song Shin, 2004, Does one Soros make a difference? a theory of currency crises with large and small traders, *Review of Economic Studies* 71, 87–113.
- DeLong, J. Bradford, Andrei Shleifer, Lawrence Summers, and Robert J. Waldmann, 1990, Noise trader risk in financial markets, *Journal of Political Economy* 98, 703–738.
- Dow, James, and Gary Gorton, 1994, Arbitrage chains, *Journal of Finance* 49, 819–849.
- , 1997, Noise trading, delegated portfolio management, and economic welfare, *Journal of Political Economy* 105, 1024–1050.
- Fama, Eugene F., and Kenneth R. French, 1993, Common risk factors in the returns on stocks and bonds, *Journal of Financial Economics* 33, 3–56.
- Fishman, Michael, and Kathleen Hagerty, 1995, The mandatory disclosure of trades and market liquidity, *Review of Financial Studies* 8, 637–676.
- Froot, Kenneth, David Scharfstein, and Jeremy Stein, 1992, Herd on the street: Information inefficiencies in a market with short-term speculation, *Journal of Finance* 47, 1461–1484.
- Goldman, Eitan, and Steve L. Slezak, 2003, Delegated portfolio management and rational prolonged mispricing, *Journal of Finance* 58, 283–311.

- Huddart, Steven, John S. Hughes, and Carolyn B. Levine, 2001, Public disclosure and dissimulation of insider trades, *Econometrica* 69, 665–681.
- John, Kose, and Ranga Narayanan, 1997, Market manipulation and the role of insider trading regulations, *Journal of Business* 70, 217–247.
- Kyle, Albert S., 1985, Continuous auctions and insider trading, *Econometrica* 53, 1315–1335.
- Prendergast, Canice., and Lars Stole, 1996, Impetuous youngsters and jaded oldtimers: An analysis of behavioral decision-making rules, *Journal of Political Economy* 104, 1105–1134.
- Shleifer, Andrei, and Robert W. Vishny, 1990, Equilibrium short horizons of investors and firms, *American Economic Review Papers and Proceedings* 80, 148–153.
- Slater, Robert, 1996, *Soros: The Unauthorized Biography* (McGraw-Hill: New York).
- Sobel, Joel, 1985, A theory of credibility, *Review of Economic Studies* 52, 557–573.

A Derivation of Equilibrium Short-Run Profits

This appendix derives the equilibrium short-run profits presented in section 3.3 and used throughout the paper. Market makers set prices in the first period under the assumption that all order flow comes from either the passive investor or noise traders. The ex ante probability of the leader trading in a given asset is zero. The leader's trades impact the price, but he effectively escapes notice until he makes a disclosure regarding an asset. Market makers base their expected value of the asset on their conjecture of the investor's trading strategy. We solve for equilibrium between market makers and passive investors just as is done in Section I.C of Froot, Scharfstein, and Stein (1992).

A.1 Market Makers

The observed order flow at time t , F_t , for $t = 1, 2$, can be written:

$$F_1 = Q_{1L} + \frac{1}{2}Q_I + u_1 \quad (22)$$

$$F_2 = Q_{2L} + \frac{1}{2}Q_I + u_2 \quad (23)$$

where Q_{1L}, Q_{2L} are the equilibrium demands of the leader and Q_I is the investor's demand. Equilibrium demands are a function of each trader's private information. The assumption of forced early liquidation parameterized by α means here that the component of time 2 order flow coming from the leader, Q_{2L} , is equal to $-Q_{1L}$ with probability $1 - \alpha$. We will assume for our main results that the leader does not trade if he is not forced into early liquidation, or that Q_{2L} is equal to zero. This assumption will be incorporated below, although we first derive the price equations with a general Q_{2L} . We will use the general expression in section B.1, where we permit the leader to choose second period demand optimally whenever he is not forced to liquidate.

Given that market makers' priors are normally distributed around a mean of zero, their posterior belief having seen the date-1 order flow, F_1 , is just F_1 multiplied by some constant, μ_1 . This constant is equal to the coefficient in a regression of $\tilde{v}$ on F_1 , taking Q_{1L} to be negligible. Thus, the price at time 1, P_1 , equals μF_1 , where

$$\mu_1 = \frac{\text{cov}^{MM}(\tilde{v}, F_1)}{\text{var}^{MM}(F_1)} = \frac{\text{cov}(\tilde{v}, \frac{1}{2}Q_I + \tilde{u}_1)}{\text{var}(\frac{1}{2}Q_I + \tilde{u}_1)} \quad (24)$$

Similarly, the order flow in the second period conveys information about the fundamental value. Although market makers could potentially use F_1 and F_2 to gather additional information regarding whether the leader is honest or deceptive via inference on Q_{1L} and Q_{2L} , its relative importance is only very slight if σ^2 is relatively high. If the ratio $\frac{\eta}{\sigma}$ is kept constant, increasing both η^2 and σ^2 need not change the information content of F_1 or F_2 regarding $\tilde{b}$ (and hence the market depth). We will presume for tractability that η^2 and σ^2 are much higher than the difference in Q_{1L} for honest and deceptive disclosures, so that market makers consider order flows to only be informative regarding $\tilde{b}$. This is most naturally interpreted as presuming that background informed trading on $\tilde{b}$ is the primary component of trading volume.

Since $\tilde{a}$ and $\tilde{b}$ are independent, the price at time 2, P_2 , is the sum of the updated expectation of each component of $\tilde{v}$ separately. We assumed in the text that market makers know whether or not the leader is forced into early liquidation. Suppose first that the leader is forced into early liquidation. Thus his demand at time 2 exactly cancels his earlier demand. By combining order flows from periods 1 and 2, market makers get a more precise signal of $\tilde{b}$ with less extraneous noise from the leader's trades. Since the variances of $\tilde{u}_1$ and $\tilde{u}_2$ are the same, market makers put equal weight on the order flows in each period in forming their expectations about the asset's value at time 2. The combined signal, $(F_1 + F_2)/2$, is normally distributed and is not affected by changes in Q_{1L} . Thus, the market maker's conditional expectation of $\tilde{b}$ is some constant μ_2 multiplied by average order flow. Just as in the first period, this constant is the coefficient in a regression of $\tilde{b}$ on $(F_1 + F_2)/2$, or

$$\mu_2 = \frac{\text{cov}\left(\tilde{b}, \frac{1}{2}Q_I + \frac{\tilde{u}_1 + \tilde{u}_2}{2}\right)}{\text{var}\left(\frac{1}{2}Q_I + \frac{\tilde{u}_1 + \tilde{u}_2}{2}\right)} \quad (25)$$

The constants μ_1 and μ_2 above are market depth parameters. Since the information about $\tilde{v}$ in the combined order flow is more precise than the information in the first period order flow, it follows that $\mu_2 > \mu_1$. Of course, the equilibrium values of μ_1 and μ_2 depend on the trading strategy of the passive investor, and this trading strategy, in turn, depends on the way in which the market maker sets prices.

Now suppose the leader is not forced into early liquidation. If the difference between the demands of honest and deceptive leaders is small and insignificant relative to the common component of demand, as is the case for low α , then we may treat the difference between average order flow $(F_1 + F_2)/2$ and expected leader demand $E\left[\frac{Q_{1L} + Q_{2L}}{2}\right]$ as a signal for $\tilde{b}$ just as $(F_1 + F_2)/2$ alone was used in the case of forced liquidation. In other words, market makers multiply this difference by the same market depth constant μ_2 defined in (25) to obtain a more precise estimate of $\tilde{b}$. Thus regardless of whether or not early liquidation occurs the price at time 2 is

$$P_2 = E[\tilde{a}] + \frac{\mu_2}{2}(F_1 + F_2 - E[Q_{1L} + Q_{2L}]) \quad (26)$$

At time 2, the only new information market makers receive regarding $\tilde{a}$ is the message $\tilde{m} = 1$. However, they are unsure whether this message was sent honestly or deceptively. As mentioned in the text, we denote the ex ante probability a message was sent honestly with π , the credibility. Even though honest and deceptive leaders choose different demands, market makers do not update the probability of honesty because of the above assumption regarding noise.¹⁹ The updated probability that $\tilde{a} = 1$ following a disclosure is $\pi p + (1 - \pi)\frac{1}{2}$. Hence the market maker's expectation of $\tilde{a}$ following disclosure is $E[\tilde{a}|\tilde{m} = 1] = \pi(2p - 1)$. Note that the updated expectation is increasing in both π and p .

Given the updated expectations for $\tilde{a}$ and $\tilde{b}$, the market maker sets

$$P_2^{1-\alpha} = \pi(2p - 1) + \mu_2\left(\frac{F_1 + F_2}{2}\right) \quad (27)$$

¹⁹If instead there were relatively little noise, market makers would have to update π based on order flow and deceptive leaders would be forced to partially mimic the demand of honest leaders by raising their demands. Solving such a model is further complicated by the possibility of pooled and separating equilibria.

if the leader is forced to liquidate early. If the leader is not liquidated early, then the price at date 2 will be

$$P_2^\alpha = \pi(2p - 1) + \mu_2 \left(\frac{F_1 + F_2}{2} - E \left[\frac{Q_{1L} + Q_{2L}}{2} \right] \right) \quad (28)$$

where

$$E[Q_{1L} + Q_{2L}] = \pi Q_{1L}^H + (1 - \pi) Q_{1L}^D + \pi Q_{2L}^H + (1 - \pi) Q_{2L}^D. \quad (29)$$

The latter price at date 2, P_2^α , for when the leader is not forced into early liquidation, will only be used in the extension in section B.1 below.

The change in price from period 1 to 2 can be attributed to two distinct factors. The first is the change in the expectation of $\tilde{a}$ from zero to $\pi(2p - 1)$. Given $\tilde{s} = 1$, this change is always nonnegative for any leader that the public believes has a signal that is at least worthless, i.e. $p \geq \frac{1}{2}$. The second change in price comes from the revision in the expectation of $\tilde{b}$. This revision is typically a downward revision since F_1 contained the positive term Q_{1L} but the new signal for $\tilde{b}$ does not. Put another way, the price declines because not as much of previous order flow came from speculation on $\tilde{b}$ as was previously thought. This decline, however, comes about regardless of disclosure. Therefore it is easy to see how disclosure is always optimal for an informed leader. Once again, this result will change when we allow for trading in the second period.

A.2 The Passive Investor

The trading strategy of the passive investor depends on the information she observes, namely the realized value b of $\tilde{b}$. In forming her demand, she considers the probability the leader will get involved in this risky asset to be negligible ex ante, and will ignore his effect on prices. She expects to transact at an average entry price of $(P_1 + P_2)/2$ and exit at $\tilde{v}$. Thus her expected profit on each unit purchased is $E[\tilde{v} - (P_1 + P_2)/2 | \tilde{b} = b]$, and so her expected utility is $U_I = Q_I E[\tilde{v} - (P_1 + P_2)/2 | \tilde{b} = b]$.

The expectation of $\tilde{v}$ for a passive investor who has observed b is just b . Observing b also enables her to forecast prices P_1 and P_2 , since she knows how market makers will react to her demand. However, she knows nothing of the order flow generated by the noise traders, and these flows have zero mean to her. Taking the pricing strategy of the market maker (in particular, μ_t) as given, her expectation of the price at time t is $E[P_t | b] = \mu_t E[\frac{1}{t} \sum F_t | b] = \mu_t Q_I/2$. Given these expectations, the optimal demand Q_S conditional on observing b maximizes

$$U_I = Q_I \left(b - \frac{1}{4} (\mu_1 + \mu_2) Q_I \right). \quad (30)$$

The first order condition for Q_S yields $Q_S^* = \frac{2b}{\mu_1 + \mu_2}$. Applying what we know of the unconditional distribution of $\tilde{b}$, we find that the variance of $\frac{1}{2} Q_S$ is $\frac{\eta^2}{(\mu_1 + \mu_2)^2}$ and the covariance of $\tilde{b}$ with $\frac{1}{2} Q_S$ is $\frac{\eta^2}{\mu_1 + \mu_2}$. Next,

use these expressions in (24) and (25) to find the system of equations for μ_1 and μ_2 :

$$\mu_1 = \frac{\frac{\eta^2}{\mu_1 + \mu_2}}{\left(\frac{\eta}{\mu_1 + \mu_2}\right)^2 + \sigma^2} \quad (31a)$$

$$\mu_2 = \frac{\frac{\eta^2}{\mu_1 + \mu_2}}{\left(\frac{\eta}{\mu_1 + \mu_2}\right)^2 + \frac{1}{2}\sigma^2} \quad (31b)$$

Solving this system of equations, we find

$$\mu_1 = \left(\frac{\sqrt[4]{2}}{1 + \sqrt{2}}\right) \frac{\eta}{\sigma} \quad (32a)$$

$$\mu_2 = \left(\frac{\sqrt[4]{2}}{1 + \frac{1}{2}\sqrt{2}}\right) \frac{\eta}{\sigma} \quad (32b)$$

Note from equations (31) that $\mu_2 > \mu_1$ as we claimed earlier. Also note that $\mu_1 + \mu_2 = \sqrt[4]{2} \frac{\eta}{\sigma}$. The optimal demand is

$$Q_I^* = 2^{3/4} \frac{\sigma}{\eta} b \quad (33)$$

and so ex ante expected utility is

$$\eta\sigma \left(\sqrt{2} - \frac{1}{\sqrt[4]{2}} \right) \quad (34)$$

A.3 Leader Demand and Trading Profits

The leader's demand depends on the information he observes and whether or not he will choose to disclose his opinion. If the leader receives the signal $\tilde{s}$, he makes either an honest disclosure or no disclosure. On the other hand, a leader without any signal will never enter into a position in the risky asset without intending to make a deceptive disclosure.

In forming his demand, the leader takes the market maker's equilibrium pricing strategy (i.e. the values of the market depth parameters μ_1 and μ_2 in (32)) as given. Consider the demand decision facing the leader in the first period before making a disclosure. If the leader is informed ($\tilde{s} = 1$), he makes an honest disclosure. If the leader is uninformed ($\tilde{s} = \emptyset$), he makes a deceptive disclosure. With probability α , whatever quantity the leader buys in period 1 at P_1 he is able to sell at the full information value $\tilde{v}$. With probability $1 - \alpha$, he is forced to liquidate at time 2 at the prevailing price, $P_2^{1-\alpha}$. Hence expected profits on each unit purchased are $E[\alpha\tilde{v} + (1 - \alpha)P_2^{1-\alpha} - P_1 | Q_{1L}, \tilde{s}]$. The leader's signal is not informative of the order flow generated by either noise traders or the passive investor, these flows have zero mean to him. Furthermore, the leader knows that if he is forced into liquidation at time 2, his demand at time 2 will exactly cancel his earlier demand at time 1, and so expected total order flow $F_1 + F_2$ is zero as well. Thus $E[P_1 | Q_L]$ is $\mu_1 Q_{1L}$ and $E[P_2^{1-\alpha} | Q_{1L}]$ is $\pi(2p - 1)$. Finally, the expectation of $\tilde{v}$ conditional on observing $\tilde{s} = 1$ is $2p - 1$, whereas if the leader does not receive a signal his expectation of $\tilde{v}$ is 0. Accounting also for

fixed disclosure costs of k , the leader's expected utility is

$$U(Q_{1L}, \tilde{s}; \pi) = \begin{cases} Q_{1L}((2p-1)(\alpha + (1-\alpha)\pi) - \mu_1 Q_{1L}) - k & \text{if } \tilde{s} = 1 \\ Q_{1L}((2p-1)(1-\alpha)\pi - \mu_1 Q_{1L}) - k & \text{if } \tilde{s} = \emptyset \end{cases}. \quad (35)$$

The leader chooses Q_{1L} to maximize $U(Q_{1L}, \tilde{s})$. The first order condition for demand yields

$$Q_{1L}^* = \begin{cases} \left(\frac{1+\sqrt{2}}{2^{3/2}}\right) \frac{\sigma}{\eta} (2p-1)(\alpha + (1-\alpha)\pi) & \text{if } \tilde{s} = 1 \\ \left(\frac{1+\sqrt{2}}{2^{3/2}}\right) \frac{\sigma}{\eta} (2p-1)(1-\alpha)\pi & \text{if } \tilde{s} = \emptyset \end{cases} \quad (36)$$

When the leader is informed, demand is increasing in α because the public does not have as much confidence in the leader's message as the leader himself does (i.e. $\pi < 1$). Conversely, demand is decreasing in α when the leader is uninformed because the public has too much confidence in the message ($\pi > 0$). Note that the difference between demands for $\tilde{s} = 1$ and $\tilde{s} = \emptyset$ goes to zero as the liquidation risk becomes very high ($\alpha \rightarrow 0$). Thus our assumption that market makers do not make use of order flows to infer additional information regarding the honest disclosure probability is not very restrictive for small α . Likewise, our use of $\frac{F_1+F_2}{2} - E\left[\frac{Q_{1L}+Q_{2L}}{2}\right]$ as a signal for $\tilde{b}$ is accurate for small α . Also note that if the leader has a poor reputation ($\rho \rightarrow 0$, which implies $\pi \rightarrow 0$), his demand approaches zero as his early liquidation risk rises. This parallels the result of Froot, Scharfstein, and Stein (1992), who show that it is not possible for traders with short trading horizons to profit from their information if they are the only ones in the market with that information. See section 3.4 for a discussion of further similar results for when the leader is allowed to choose between learning $\tilde{s}$ and $\tilde{b}$.

Given the optimal demand functions above, the expected utility can be summarized as a function of only the signal $\tilde{s}$ and the credibility π :

$$U(\tilde{s}, \pi) = \begin{cases} \lambda(2p-1)^2(\alpha + (1-\alpha)\pi)^2 - k \equiv U_H(\pi) & \text{if } \tilde{s} = 1 \\ \lambda(2p-1)^2(1-\alpha)^2\pi^2 - k \equiv U_D(\pi) & \text{if } \tilde{s} = \emptyset \end{cases}. \quad (37)$$

where $\lambda \equiv \frac{1}{4\mu_1} = \left(\frac{1+\sqrt{2}}{4^{3/2}}\right) \frac{\sigma}{\eta}$ is a market liquidity parameter directly related to $\frac{\sigma}{\eta}$.

Finally we consider whether the leader decides to send any message. If no message is sent, market makers are unaware of the leader's presence in the market; both the initial trade and the unravelling trade are treated as if they were demand from the passive investor and noise traders. In particular, prices are set as $P_1 = \mu_1 F_1$ at time 1 and $P_2 = \mu_2 \left(\frac{F_1+F_2}{2}\right)$ at time 2. Thus the leader's expected price is $\mu_1 Q_{1L}$ at time 1 and 0 at time 2 when forced into liquidation. Because of the adverse price impact from establishing a position and the expected liquidation value of 0, a leader without a signal will never enter the market without intending to make a deceptive disclosure. For an informed leader, however, the expectation of $\tilde{v}$ conditional on observing $\tilde{s} = 1$ is $2p-1$. Hence his expected utility is (the subscript T denotes trade without disclosure)

$$U_T = Q_{1L}((2p-1)\alpha - \mu_1 Q_{1L}).$$

Substituting the first order condition for demand, $Q_{1L} = \left(\frac{1+\sqrt{2}}{2^{3/2}}\right) \frac{\sigma}{\eta} (2p-1) \alpha$, we obtain

$$U_T = \lambda (2p-1)^2 \alpha^2 \quad (38)$$

Only if the fixed disclosure cost k is relatively high and credibility π is very low will an informed leader choose to trade on his information without disclosing it. In the extension below, we allow the leader to trade again in the second period if he is not forced into liquidation. As a result, an informed leader chooses not to disclose for an even greater range of credibilities.

B Extensions

B.1 Post-Announcement Trading

Speculators who let others know their position face numerous risks, such as the risk of predatory trading should they receive a margin call (see e.g. Brunnermeier and Pederson (2005)), or the risk that new information will come out that causes them to want to increase their position. While these risks are not the focus of this paper, the mechanism we study may explain why a large trader would voluntarily disclose his positions. An extension of the model discusses when large traders will choose to keep their presence a secret. We show that traders keep their best information to themselves. For any disclosure to take place, their reputation for being informed must be sufficiently high or they must have a very high probability of early liquidation. In this section we return to the single round trading model from section 3 and expand it to account for trading after the leader sends a message. The section builds directly on the notation and results from Appendix A above.

B.1.1 Second Period Demand

Noise traders play a particularly important role in this extension. In addition to the standard function of hiding the investor's informative trades (thus limiting price impact and allowing informed traders to profit), the noise also masks the leader's trades, preventing the public from learning the leader's true position size. The size of the leader's position turns out to be much more important when the leader can trade after the announcement.

In forming his demand, the leader takes the market maker's equilibrium pricing strategy (i.e. the values of the market depth parameters μ_1 and μ_2 , equation (32)) as given. As is standard in such problems, begin by solving for the second period demand first. If the leader is forced to liquidate early, this demand is $-Q_{1L}$ by necessity. Suppose then that the leader is allowed to optimize his trade in the second period. The leader must form an expectation for the price at which he will trade in the second period. Whether or not he had chosen to disclose earlier, this price depends linearly on the term F_2 , which the leader estimates using the realized values of F_1 and Q_{1L} , and his own projected demand Q_{2L} . Not only does the leader recognize that some of the order flow, past and present, comes from him, but he also has an improved estimation of $\tilde{b}$ relative to market makers. His expectation of $\tilde{b}$ given that, in fact, only $F_1 - Q_{1L}$ of first period order flow

came from investors and noise traders, is some constant multiplied by $F_1 - Q_{1L}$, where the constant is the following regression coefficient:

$$\frac{\text{cov}\left(\tilde{v}, 2^{-1/4} \frac{\sigma}{\eta} \tilde{b} + \tilde{u}_1\right)}{\text{var}\left(2^{-1/4} \frac{\sigma}{\eta} \tilde{b} + \tilde{u}_1\right)} = \left(\frac{\sqrt{2}}{1 + \eta^2 \sqrt{2}}\right) \frac{\eta}{\sigma} \quad (39)$$

While the leader may have no information regarding noise trader demand, his improved estimate of $\tilde{b}$ allows him to better estimate the $\frac{1}{2}Q_I$ component of F_2 . Applying the updated estimate of $\tilde{b}$ in (33) we find the leader's expectation of time 2 order flow is

$$\frac{2^{1/4}}{1 + \eta^2 \sqrt{2}} (F_1 - Q_{1L}) + Q_{2L} \quad (40)$$

The leader may decide his retrade demand after either disclosure or nondisclosure. Suppose first that the leader did not announce any message $\tilde{m}$ in the first period, so his second period demand is treated as if it were demand from the passive investor. In particular, the price is set as $P_2 = \frac{\mu_2}{2} (F_1 + F_2)$. The expectation of $\tilde{v}$ conditional on observing $\tilde{s} = 1$ is $2p - 1$ (a deceptive leader will always announce, so we need only consider the incentives of an honest leader). Hence the leader's expected utility in period 2 is

$$U_{2L}^N = Q_{2L} \left(2p - 1 - \frac{\mu_2}{2} \left(\left(1 + \frac{2^{1/4}}{1 + \eta^2 \sqrt{2}} \right) (F_1 - Q_{1L}) + Q_{1L} + Q_{2L} \right) \right), \quad (41)$$

which is maximized at

$$Q_{2L}^N = \frac{2p - 1}{\mu_2} - \frac{1}{2} \left(\left(1 + \frac{2^{1/4}}{1 + \eta^2 \sqrt{2}} \right) (F_1 - Q_{1L}) + Q_{1L} \right) \quad (42)$$

Note that the leader's expectation of Q_{2L} at time 1 is $\frac{2p-1}{\mu_2} - \frac{Q_{1L}}{2}$. We have denoted equilibrium values for a non-disclosing leader

We will denote equilibrium values for leaders making honest or deceptive disclosures with superscripts H and D , respectively. Equilibrium values for a non-disclosing leader (who possesses a signal) are denoted by a superscript N .

Now suppose that the leader announces the message $\tilde{m} = 1$ in the first period, so the price in the second period is set as in (28). The $\frac{F_1 + F_2}{2}$ term in P_2^α is evaluated exactly as above, but now there is an additional term (29). Because of this term, honest and deceptive leaders take slightly different derivatives when maximizing $U_{L-R} = Q_{2L} (E[\tilde{v}|\tilde{s}] - E[P_2^\alpha])$.²⁰ For an honest leader, U_{2L}^H is the product of Q_{2L}^H and

$$(1 - \pi)(2p - 1) - \frac{\mu_2}{2} E[F_1 + F_2 - E[Q_{1L} + Q_{2L}] | I_2, \tilde{s}], \quad (43)$$

where I_2 denotes any leader's information set at time 2. This set includes the past order flow, F_1 , since

²⁰This section must be modified at some point. It turns out that $Q_{2D}^H = -Q_{2L}^D$, and both are far from zero, so that market makers can easily detect deception in the second period. One possibility is to assume that market makers are watching closely now, so deceptive leaders must mimic honest ones. This does not make perfect sense, however, since the deceptive leader would rather not trade at all and let prices revert.

observing order flow is equivalent to observing the past price given knowledge of the market maker's pricing strategy. Past and current demands, Q_{1L} and Q_{2L} , are included in I_2 as well. Since an honest leader is truly informed, his total information set also includes $\tilde{s}$. For a deceptive leader, U_{2L}^D is the product of Q_{2L}^D and

$$-\pi(2p-1) - \frac{\mu_2}{2} E[F_1 + F_2 - E[Q_{1L} + Q_{2L}] | I_2] \quad (44)$$

Since the signal $\tilde{s}$ is not informative regarding order flows, $E[F_1 + F_2 - E[Q_{1L} + Q_{2L}] | I_2] =$

$$\left(1 + \frac{2^{1/4}}{1+\eta^2\sqrt{2}}\right) (F_1 - Q_{1L}) + (1-\pi)(Q_{2L}^H - Q_{2L}^D) + (1-\pi)(Q_{1L}^H - Q_L^D)$$

for an honest leader and

$$\left(1 + \frac{2^{1/4}}{1+\eta^2\sqrt{2}}\right) (F_1 - Q_{1L}) - \pi(Q_{1L}^H - Q_L^D) - \pi(Q_{2L}^H - Q_{2L}^D) \quad (45)$$

for a deceptive one. We deliberately leave the superscript off of Q_L in the term $F_1 - Q_{1L}$, since it is, in fact, an “unexpected order flow” term of $\frac{1}{2}\tilde{Q}_S + \tilde{u}_1$. This term has zero mean with respect to only time 1 information but is known at time 2. Honest and deceptive leaders solve for Q_{2L}^H and Q_{2L}^D , respectively, as

$$Q_{2L}^H = \frac{2p-1}{\mu_2} - \frac{1}{2} \left(Q_{1L}^H + \left(1 + \frac{2^{1/4}}{1+\eta^2\sqrt{2}}\right) \frac{F_1 - Q_{1L}}{1-\pi} - Q_{1L}^D - Q_{2L}^D \right)$$

and

$$Q_{2L}^D = -\frac{2p-1}{\mu_2} - \frac{1}{2} \left(Q_{1L}^D + \left(1 + \frac{2^{1/4}}{1+\eta^2\sqrt{2}}\right) \frac{F_1 - Q_{1L}}{\pi} - Q_{1L}^H - Q_{2L}^H \right)$$

These equations constitute a system of equations for Q_{2L}^H and Q_{2L}^D with solution

$$\begin{aligned} Q_{2L}^H &= \frac{2}{3} \left(\frac{2p-1}{\mu_2} \right) - \frac{1}{3} (Q_{1L}^H - Q_{1L}^D) - \left(1 + \frac{2^{1/4}}{1+\eta^2\sqrt{2}}\right) \left(\frac{1+\pi}{3\pi(1-\pi)} \right) (F_1 - Q_{1L}) \\ Q_{2L}^D &= -\frac{2}{3} \left(\frac{2p-1}{\mu_2} \right) + \frac{1}{3} (Q_{1L}^H - Q_{1L}^D) - \left(1 + \frac{2^{1/4}}{1+\eta^2\sqrt{2}}\right) \left(\frac{2-\pi}{3\pi(1-\pi)} \right) (F_1 - Q_{1L}) \end{aligned}$$

Hence the Q_{2L} values chosen by honest and deceptive leaders are exactly the opposite of each other, aside from slightly different slope terms multiplying the unexpected order flow, $F_1 - Q_{1L}$. Now we can simplify the term $E[F_1 + F_2 - E[Q_{1L} + Q_{2L}] | I_2]$. For an honest leader, it is

$$\left(1 + \frac{2^{1/4}}{1+\eta^2\sqrt{2}}\right) \left(\frac{1+\pi}{3\pi} \right) (F_1 - Q_{1L}) + \frac{4(1-\pi)}{3} \left(\frac{2p-1}{\mu_2} \right) + \frac{(1-\pi)}{3} (Q_{1L}^H - Q_{1L}^D)$$

and for a deceptive leader,

$$\left(1 + \frac{2^{1/4}}{1+\eta^2\sqrt{2}}\right) \left(\frac{2-\pi}{3(1-\pi)} \right) (F_1 - Q_{1L}) - \frac{4\pi}{3} \left(\frac{2p-1}{\mu_2} \right) - \frac{\pi}{3} (Q_{1L}^H - Q_{1L}^D)$$

B.1.2 Initial Demand

Next, consider the time 1 decision facing a leader after observing $\tilde{s} = 1$ intending to make an honest disclosure. With probability α , whatever quantity the leader buys in period 1 at P_1 he is able to sell at

the full information value $\tilde{v}$, with an opportunity to earn additional utility U_{2L} from retrade in the second period. With probability $1 - \alpha$, he is forced to liquidate at time 2 at the prevailing price, $P_2^{1-\alpha}$. Hence total utility is $U_L + \alpha U_{2L}$, where U_L is utility originating only from the initial demand. Expected profits on each unit purchased are $E[\alpha \tilde{v} + (1 - \alpha) P_2^{1-\alpha} - P_1 | I_1, \tilde{s}]$, where I_1 denotes the leader's information set at time 1, which includes his own demand Q_L and knowledge of whether or not he intends to disclose. The leader's information is not informative of the order flow generated by either noise traders or the passive investor, these flows have zero mean to him. Furthermore, the leader knows that if he is forced into liquidation at time 2, his demand at time 2 will exactly cancel his earlier demand at time 1, and so expected total order flow is zero as well. Thus $E[P_1 | I_1]$ is $\mu_1 Q_L$ and $E[P_2^{1-\alpha} | I_1]$ is $\pi(2p - 1)$. Finally, the expectation of $\tilde{v}$ conditional on observing $\tilde{s} = 1$ is $2p - 1$. Hence the honest leader's expected utility from Q_L alone is

$$U_L^H = Q_{1L} ((2p - 1)(\alpha + (1 - \alpha)\pi) - \mu_1 Q_{1L}). \quad (46)$$

Likewise, a deceptive leader goes through a similar calculation, except he does not know the value of signal $\tilde{s}$, and so his estimate of $\tilde{v}$ is zero. The expectations for P_1 and $P_2^{1-\alpha}$ do not depend on $\tilde{s}$, hence

$$U_L^D = Q_{1L} ((2p - 1)(1 - \alpha)\pi - \mu_1 Q_{1L}) \quad (47)$$

The leader chooses Q_L to maximize $U_L + \alpha E[U_{2L}]$. The functional form of $E[U_{2L}]$ is rather involved, so we differentiate under the integral sign, rather than evaluating the expectation first. Since we are differentiating with respect to the known variable Q_L , and the integral is an expectation where $\tilde{b}$ and $\tilde{u}_1$ are the only sources of randomness, the derivative can be taken inside the expectation directly. One easily finds that $\frac{dQ_{2L}^H}{dQ_{1L}^H} = \frac{dQ_{2L}^D}{dQ_{1L}^D} = -\frac{1}{3}$ and $\frac{dQ_{2L}^D}{dQ_{1L}^H} = \frac{dQ_{2L}^H}{dQ_{1L}^D} = \frac{1}{3}$. For now we take as given that $\frac{dQ_{1L}^D}{dQ_{1L}^H} = \frac{dQ_{1L}^H}{dQ_{1L}^D} = 0$, as will later be verified. Intermediate steps:

$$\begin{aligned} \frac{dU_{2L}^H}{dQ_{1L}^H} &= \frac{dQ_{2L}^H}{dQ_{1L}^H} \left(\frac{U_{2L}^H}{Q_{2L}^H} \right) - \frac{\mu_2}{2} (1 - \pi) \left(1 + \frac{dQ_{2L}^H}{dQ_{1L}^H} - \frac{dQ_{2L}^D}{dQ_{1L}^H} \right) Q_{2L}^H \\ &= -\frac{1}{3} \left(\frac{U_{2L}^H}{Q_{2L}^H} \right) - \frac{\mu_2}{6} (1 - \pi) Q_{2L}^H \\ E \left[\frac{dU_{2L}^H}{dQ_{1L}^H} \middle| I_2, \tilde{s} \right] &= -\frac{1}{3} (1 - \pi) \left((2p - 1) - \frac{\mu_2}{2} (Q_{1L}^H - Q_{1L}^D + Q_{2L}^H - Q_{2L}^D) \right) \\ &\quad - \frac{\mu_2}{6} (1 - \pi) \left(\frac{2}{3} \left(\frac{2p - 1}{\mu_2} \right) - \frac{1}{3} (Q_{1L}^H - Q_{1L}^D) \right) \end{aligned}$$

Thus $E \left[\frac{dU_{2L}^H}{dQ_{1L}^H} \middle| I_2, \tilde{s} \right]$ equals

$$- (1 - \pi) \frac{2}{9} (2p - 1) + \frac{\mu_2}{9} (1 - \pi) (Q_{1L}^H - Q_{1L}^D) \quad (48)$$

Now for deceptive leaders

$$\frac{dU_{2L}^D}{dQ_{1L}^D} = -\frac{1}{3} \left(\frac{U_{2L}^D}{Q_{2L}^D} \right) - \frac{\mu_2}{6} \pi Q_{2L}^D$$

$$E \left[\frac{dU_{2L}^D}{dQ_{1L}^D} \middle| I_2 \right] = \frac{1}{3} \pi \left((2p-1) - \frac{\mu_2}{2} (Q_{1L}^H - Q_{1L}^D + Q_{2L}^H - Q_{2L}^D) \right) - \frac{\mu_2}{6} \pi \left(-\frac{2}{3} \left(\frac{2p-1}{\mu_2} \right) + \frac{1}{3} (Q_{1L}^H - Q_{1L}^D) \right)$$

Thus $E \left[\frac{dU_{2L}^D}{dQ_{1L}^D} \middle| I_2 \right]$ equals

$$\pi \frac{2}{9} (2p-1) - \frac{\mu_2}{9} \pi (Q_{1L}^H - Q_{1L}^D) \quad (49)$$

The honest leader's total first order condition for maximizing $\alpha E [U_{L-R}^H] + U_L^H$ with respect to Q_{1L}^H is

$$-(1-\pi) \frac{2}{9} (2p-1) \alpha + \frac{\mu_2}{9} (1-\pi) (Q_{1L}^H - Q_{1L}^D) \alpha + (2p-1) (\alpha + (1-\alpha) \pi) = 2\mu_1 Q_{1L}^H \quad (50)$$

and for the deceptive leader

$$\pi \frac{2}{9} (2p-1) \alpha - \frac{\mu_2}{9} \pi (Q_{1L}^H - Q_{1L}^D) \alpha + (2p-1) (1-\alpha) \pi = 2\mu_1 Q_{1L}^D \quad (51)$$

Solving the system of equations for $Q_{1L}^H - Q_{1L}^D$:

$$Q_{1L}^H - Q_{1L}^D = \frac{7\alpha(2p-1)}{18\mu_1 - \alpha\mu_2} \equiv \Lambda(2p-1) \quad (52)$$

Thus we find

$$Q_{1L}^H = \frac{2p-1}{2\mu_1} (\alpha + \pi(1-\alpha) - (1-\pi)\alpha\Delta) \quad (53)$$

$$Q_{1L}^D = \frac{2p-1}{2\mu_1} (\pi(1-\alpha) + \pi\alpha\Delta) \quad (54)$$

The sign of $\Delta \equiv \frac{1}{9} (2 - \mu_2 \Lambda) =$

$$\frac{4\mu_1 - \alpha\mu_2}{18\mu_1 - \alpha\mu_2} \quad (55)$$

is positive, as it can be shown rather easily that $4\mu_1 > \mu_2$. Note that the difference between Q_{1L}^H and Q_{1L}^D goes to zero as the liquidation risk increases ($\alpha \rightarrow 0$). Therefore our assumption that market makers do not make use of order flows to infer additional information regarding the honest disclosure probability is justified for small α . Furthermore, both Q_{1L}^H and Q_{1L}^D are functions exclusively of parameters of the one trading round model, thus justifying our earlier assumption that $\frac{dQ_{1L}^D}{dQ_{1L}^H} = \frac{dQ_{1L}^H}{dQ_{1L}^D} = 0$.

B.1.3 Disclosure Optimality Condition

Now we can solve for the informed leader's optimality condition regarding disclosure. Not announcing yields a return to units purchased in the first period only of

$$E \left[\alpha \tilde{v} + (1-\alpha) \frac{\mu_2}{2} (F_1 + F_2) - \mu_1 F_1 \middle| I_1, \tilde{s} \right].$$

The sum of order flows has zero mean as of time 1. As before, $E[\tilde{v}|\tilde{s}] = (2p-1)$ and $E[F_1|I_1] = Q_{1L}$, hence this return is just $\alpha(2p-1) - \mu_1 Q_L$. In addition, the leader gets the utility $E[U_{2L}^N|I_1, \tilde{s}]$ (see (41))

with probability α . Total utility from not announcing is the sum of utility from demands in the first and second periods: $Q_{1L}(\alpha(2p-1) - \mu_1 Q_{1L}) + \alpha E[U_{2L}^N | I_1]$. Noting that $\frac{dQ_{2L}^N}{dQ_{1L}^N} = -\frac{1}{2}$, we take the derivative inside the expectation of U_{2L}^N to find the first order condition for Q_{1L} is

$$\alpha(2p-1) - 2\mu_1 Q_{1L}^N + \alpha \left(\frac{\mu_2}{4} Q_{1L}^N - \frac{(2p-1)}{2} \right) = 0 \quad (56)$$

Simplifies to

$$Q_{1L}^N = \frac{2\alpha(2p-1)}{8\mu_1 - \alpha\mu_2} \quad (57)$$

Proposition 9 *There exists some reputation ρ^* such that an informed leader discloses his signal if and only if $\rho \geq \rho^*$. Furthermore, $\rho^* \rightarrow 0$ in the limit as $\alpha \rightarrow 0$.*

The above proposition says that a good reputation is necessary in order for an informed leader to opt for disclosure. However, as the leader's liquidation risk rises, informed disclosure will not be as sensitive to reputation.

B.2 Proposed Extensions

The following two sections present alternative ways to extend the model and obtain interesting new results. These are just two examples demonstrating the versatility of the model and its potential for deriving many novel predictions and implications.

B.2.1 Variable Prediction Strength

Suppose the fundamental value $\tilde{v} = \tilde{k}\tilde{a} + \tilde{b}$, where $\tilde{k}$ is some random variable with mean one and a continuously differentiable density function on $(0, \infty)$. However, continue to assume the leader's message space is limited such that he can only convey his general "bullishness" or "bearishness," represented by $\tilde{a}$, but cannot credibly convey $\tilde{k}$ to the public, which can be interpreted as the "strength" of his prediction.²¹ Even when the leader's signal (and hence his message) is positive, the price may move down following a disclosure, precisely the opposite of the leader's intention, if the market maker's updated expectation of $\tilde{v}$ is below the previous price. Therefore it may not always be optimal for the leader to disclose his signal even when he does receive it, and for this reason the leader will choose an optimal short-run disclosure policy function $D(k, \pi) : (0, \infty) \times (0, 1) \rightarrow \{0, 1\}$, where the range of 0 or 1 denote not sending or sending the message $\tilde{m}$, respectively. If a message is not sent, all the leader's trades are lumped together with those of other traders.

Because the leader's message is sent concurrently with his demand at time 1, before price is known, the market may not respond to disclosure the way the leader hopes it does. Had we assumed he makes the disclosure decision after observing the price at time 1, he would know exactly how the public would interpret the disclosure. Thus disclosure would only take place when it benefits the leader. Our assumption that the leader must disclose before observing the period 1 price captures his lack of control regarding the effects of disclosure, and allows us to derive additional results. We could alternatively, and perhaps more realistically,

²¹Neither can he reveal the size of this position, which is equivalent.

model the additional information the public learns with a third signal, released after the leader's opinion $\tilde{m}$ is disclosed, but this would most likely make this extension intractable.²² The goal of such an extension would be to derive more refined implications regarding when honest disclosure occurs.

B.2.2 Overconfident Followers

Suppose that market makers are overconfident in the leader's information. In other words, they believe his signal precision is $\Pr(\tilde{a} = a | \tilde{s} = a) = \hat{p} > p$. Then we wish to find out whether a leader is made better or worse by overconfidence. It's easy to see how he may be better off, as he gets more of a price rise than is justified by his signal. However, if he makes the same number of deceptive disclosures as if his followers were rational, then he would make honest mistakes "too often" and would be punished for them too severely. In order to prevent his reputation from dropping towards zero, he must lower his disclosure frequency. Thus he may also be worse off because he must make fewer deceptive disclosures.

C Proofs

Proof of Proposition 1. The initial part of the proof is based on adapting Froot, Scharfstein, and Stein (1992) to our model. However, if ρ is high then the leader is able to profit from $\tilde{s}$ despite having very high early liquidation risk (low α). For very low informational content of $\tilde{b}$ (low η), this implies some point at which the leader's learning preference switches. ■

Proof of Proposition 3. Denote the charlatan's disclosure frequency by $\phi < 1$. Take initial reputation to be fixed at $\rho \in (0, 1)$. In the equations below, we suppress the argument ρ in the q , ϕ , and π functions. The public applies Bayes' rule following a nondisclosure to update the reputation, or probability of being genuine, to

$$\rho^\varnothing(\rho) \equiv \frac{(1-f)(1-q)\rho}{(1-f)(1-q)\rho + (1-\phi)(1-\rho)} \quad (58)$$

First suppose the charlatan chooses to make more disclosures than the genuine type (i.e. $\phi > f + (1-f)q$). Then reputation increases with each nondisclosure. Let $W(\rho) \in C_+$ be the discounted expected sum of the charlatan type's utility at the beginning of a period where his reputation is ρ . The charlatan chooses ϕ to maximize

$$W(\rho) = \phi \left(U_D(\pi) + \frac{\delta}{2} W(\rho^+) + \frac{\delta}{2} W(\rho^-) \right) + (1-\phi) \delta W(\rho^\varnothing) \quad (59)$$

$$= \phi U_D(\pi) - \delta \phi \left(W(\rho^\varnothing) - \frac{1}{2} (W(\rho^+) + W(\rho^-)) \right) + \delta W(\rho^\varnothing) \quad (60)$$

where the second line emphasizes gains and losses from changing disclosure frequency. By the monotonicity

²²By modelling the lack of control regarding the outcome of disclosure explicitly, we also obtain a vast reduction in complexity. Suppose the leader merely has implicit control over how his message is interpreted due to the superiority of his information, as is the case if the disclosure decision is made at the start of time 2. Then we would have to calculate the time 1 probability of choosing disclosure, since the leader sets demand to maximize the expectation of profits given that probability. This, in turn, makes market maker's inference problem vastly more complicated; they must evaluate not only how much of total order flow came from the leader and how much from other informed traders, but also what was the leader's ex ante probability of disclosure.

of W , $W(\rho) \leq W(\rho^\varnothing)$. Suppose a charlatan deviates from choosing his optimal ϕ for one period and chooses $\phi = 1$ instead. Then he will receive a utility of $U_D(\pi) + \frac{\delta}{2}(W(\rho^+) + W(\rho^-))$. Since this cannot be optimal, his value function for an optimally chosen ϕ satisfies $W(\rho) > U_D(\pi) + \frac{\delta}{2}(W(\rho^+) + W(\rho^-))$. Thus, for δ close enough to one, the immediate utility gain from disclosure of $U_D(\pi)$ is outweighed by the discounted loss of the difference between future utilities $W(\rho^\varnothing)$ and $\frac{1}{2}(W(\rho^+) + W(\rho^-))$. Hence the optimal ϕ equals the earlier established lower bound of $f + (1-f)q$ for a sufficiently high δ . At this value of ϕ , the public does not update based on nondisclosure, i.e. $\rho^\varnothing(\rho) = \rho$.

Now prove that the charlatan will not want to lower the disclosure frequency even further, below the genuine type's optimal disclosure frequency. Suppose $\delta(W(\rho) - \frac{1}{2}(W(\rho^+) + W(\rho^-))) > U_D(\pi)$, so that the charlatan wishes to lower his disclosure frequency even further. Examine the objective function for ϕ near $f + (1-f)q$. By continuity of W , strict inequalities will still hold for arbitrarily close values. Then $W(\rho) =$

$$\phi \left(U_D(\pi) - \delta \left(W(\rho) - \frac{1}{2}(W(\rho^+) + W(\rho^-)) \right) \right) + \delta w(\rho) \quad (61)$$

is the sum of a negative number and a number strictly less than $W(\rho)$, a contradiction. ■

Proofs for Section 4.1. For q in $[0, 1]$, ρ in $[0, 1]$, and W in C_+ , define

$$\rho^+(q, \rho) \equiv \left(1 - \frac{(1-\rho)f(2p-1)}{2fp + (1-f)q} \right)^{-1} \rho \quad (62)$$

$$\rho^-(q, \rho) \equiv \left(1 + \frac{(1-\rho)f(2p-1)}{2f(1-p) + (1-f)q} \right)^{-1} \rho \quad (63)$$

or, equivalently,

$$\rho^+(q, \rho) \equiv \frac{2fp + (1-f)q}{f + (1-f)q + \rho f(2p-1)} \rho \quad (64)$$

$$\rho^-(q, \rho) \equiv \frac{2f(1-p) + (1-f)q}{f + (1-f)q - \rho f(2p-1)} \rho \quad (65)$$

and define

$$V_{0D}(q; \rho, W) \equiv \lambda((1-\alpha)\pi(q, \rho))^2 - k + \frac{\delta}{2}(W[\rho^+(q, \rho)] + W[\rho^-(q, \rho)]) \quad (66)$$

$$V_{0N}(\rho, W) \equiv \delta W(\rho) \quad (67)$$

Thus

$$V_{0D}(0; \rho, W) = \lambda(1-\alpha)^2 \rho^2 - k + \frac{\delta}{2}(W[\rho^+(0, \rho)] + W[\rho^-(0, \rho)]) \quad (68)$$

$$V_{0D}(1; \rho, W) = \lambda(1-\alpha)^2 \rho^2 f^2 - k + \frac{\delta}{2}(W[\rho^+(1, \rho)] + W[\rho^-(1, \rho)]) \quad (69)$$

■

Lemma 1 *The limit of $q(\rho)$ as ρ approaches one is*

$$\begin{cases} 0 & (1-\alpha)^2 \leq \frac{k}{\lambda} \\ \frac{f}{1-f} \left((1-\alpha) \sqrt{\frac{\lambda}{k}} - 1 \right) & \frac{k}{\lambda} < (1-\alpha)^2 < \frac{k}{f^2 \lambda} \\ 1 & (1-\alpha)^2 \geq \frac{k}{f^2 \lambda} \end{cases} \quad (70)$$

Proof. In the limit as ρ approaches one, $\frac{\delta}{2} (W[\rho^+] + W[\rho^-]) - \delta W(\rho)$ goes to zero. Then the optimal q may be found by setting the instantaneous utility term $U_D(\pi)$ equal to the fixed cost k , so that $\lim_{\rho \rightarrow 1} V_{0D} - V_{0N} = 0$. Solving yields $\lim_{\rho \rightarrow 1} q(\rho) = \frac{f}{1-f} \left((1-\alpha) \sqrt{\frac{\lambda}{k}} - 1 \right)$. If this expression evaluates below (above) the unit interval, the optimal q approaches zero (one). ■

Note that the long-run honest to deceptive disclosure ratio of a genuine leader, or

$$\frac{f}{(1-f)q} = \left((1-\alpha) \sqrt{\frac{\lambda}{k}} - 1 \right)^{-1}, \quad (71)$$

does not depend on f , δ , or p . Hence we can take $f \rightarrow 0$ and $\delta \rightarrow 1$ without affecting any long-run payoff-relevant variables. For $k = 0$, only deceptive disclosures are made in the limit as $f \rightarrow 0$, whereas for any $k > 0$ there exists some $f^* > 0$ such that for all $f < f^*$ the leader voluntarily refrains from making a deceptive disclosure for any reputation.

Lemma 2 *There exists $\bar{\rho} > 0$ such that $q(\rho) = 0$ for all $\rho < \bar{\rho}$.*

Proof. To show that $q(\rho) = 0$ for some set of ρ is equivalent to showing that $V_{0N}(q; \rho, W) > V_{0D}(\rho, W)$ for all q in that set. Note that $\rho^+ - \rho = \frac{f(2p-1) - \rho f(2p-1)}{f + (1-f)q + \rho f(2p-1)}$ and $\rho - \rho^- = \frac{f(2p-1) - \rho f(2p-1)}{f + (1-f)q - \rho f(2p-1)}$, so that $\rho - \rho^- > \rho^+ - \rho$. In the neighborhood around $\rho = 0$, $W(\rho)$ is approximately linear with nonnegative slope. Since π goes to zero as ρ goes to zero for any q , $V_{0D}(q; \rho, W)$ is approximately equal to $\frac{\delta}{2} (W[\rho^+] + W[\rho^-]) - k$. Now ρ^- is further away from ρ than ρ^+ , so $\frac{\delta}{2} (W[\rho^+] + W[\rho^-])$ is strictly less than $\delta W(\rho)$, which is $V_{0N}(\rho, W)$. Taking account of the additional $-k$ term, we find that the difference between V_{0N} and V_{0D} is positive for ρ equal to some $\bar{\rho} > 0$. ■

Proof of Proposition 2. In text. ■

Proof of Proposition 4. This proof is proven in the following five steps:

Step one: V_{0D} is strictly decreasing in q on $[0, 1]$.

Sufficient condition on k and p so that V_{0D} is decreasing in q at $q = 0$, $\rho = 0$. $V_{0D}(q; \rho, W) = \lambda((1-\alpha)\pi(q, \rho))^2 - k + \frac{\delta}{2} (W[\rho^+(0, \rho)] + W[\rho^-(0, \rho)])$. Not sufficient! May be increasing in q for higher ρ .

Since π is decreasing in q (for $\rho > 0$), $U(\emptyset, \pi)$ is clearly decreasing in q as well. The derivative of V_{0D} with respect to q is

$$\frac{dV_{0D}}{dq} = -\frac{\lambda(1-\alpha)^2 2\pi^2(1-f)}{f + (1-f)q} + W'[\rho^+(q; \rho)] \frac{\partial \rho^+}{\partial q} + W'[\rho^-(q; \rho)] \frac{\partial \rho^-}{\partial q} \quad (72)$$

$$\frac{d}{dq} (W [\rho^+ (q; \rho)] + W [\rho^- (q; \rho)]) = W' [\rho^+ (q; \rho)] \frac{\partial \rho^+}{\partial q} + W' [\rho^- (q; \rho)] \frac{\partial \rho^-}{\partial q} \quad (73)$$

$$= -f (1-f) \rho (1-\rho) (2p-1) \left(\frac{W' [\rho^+ (q; \rho)]}{(f + (1-f) q + \rho f (2p-1))^2} - \frac{W' [\rho^- (q; \rho)]}{(f + (1-f) q - \rho f (2p-1))^2} \right) \quad (74)$$

Thus if the following slope condition on W holds

$$\frac{W' [\rho^+ (q; \rho)]}{W' [\rho^- (q; \rho)]} \geq \frac{(f + (1-f) q + \rho f (2p-1))^2}{(f + (1-f) q - \rho f (2p-1))^2} \quad (75)$$

then V_{0D} is decreasing in q . This condition implies that W is strictly convex. The RHS is decreasing in q , so the condition holds for all q if $q = 0$ is substituted on the RHS. We restrict consideration to equilibria such that this condition holds, and later verify that indeed it holds in the unique equilibrium we attain.

Step two: $V_{0D} (1; \rho, W) \leq V_{0N} (\rho, W)$. Suppose $V_{0D} (1; \rho, W) > V_{0N} (\rho, W)$. V_{0D} is decreasing in q and V_{0N} does not depend on q , thus by (17)

$$W (\rho) = f \left(\lambda (\alpha + (1-\alpha) \rho f)^2 + \delta (p W [\rho^+ (1; \rho)] + (1-p) W [\rho^- (1; \rho)]) \right) \quad (76)$$

$$+ (1-f) \left(\lambda (1-\alpha)^2 (\rho f)^2 + \frac{\delta}{2} (W [\rho^+ (1; \rho)] + W [\rho^- (1; \rho)]) \right) \quad (77)$$

Since $W [\rho^+ (1; \rho)] > W [\rho^- (1; \rho)]$, use $p > \frac{1}{2}$ to get

$$V_N (\rho) = \delta W (\rho) > \delta f \lambda (\alpha^2 + 2\alpha (1-\alpha) \rho f) + \delta \lambda (1-\alpha)^2 (\rho f)^2 + \frac{\delta^2}{2} (W [\rho^+] + W [\rho^-]) \quad (78)$$

$$> \delta f \lambda (\alpha^2 + 2\alpha (1-\alpha) \rho f) + \delta V_D (\rho) \quad (79)$$

As δ nears one, this must lead to a contradiction, since $f \lambda (\alpha^2 + 2\alpha (1-\alpha) \pi) > 0$ for $f > 0$. Thus $V_{0D} (1; \rho, W) \leq V_N (\rho)$.

Step three: $V_{0D} (0; \rho, W) \geq V_N (\rho)$. As $p \rightarrow \frac{1}{2}$, $W []$ expressions go to $W [\rho]$, while λ goes to zero from above. Hence if $\frac{d}{dp} \left(\frac{1}{2} (W [] + W []) \right) > 0$ (or at least greater than $-\frac{d\lambda}{dp} (1-\alpha)^2 \rho^2$), $V_{0D} (0) > V_N$ for $p > \frac{1}{2}$.

$$W \left[\frac{2p}{(1-\rho) + 2p\rho} \rho \right] > W (\rho) \quad (80)$$

$$W \left[\frac{2(1-p)}{(1-\rho) + 2(1-p)\rho} \rho \right] < W (\rho) \quad (81)$$

$$\frac{d}{dp} W \left[\frac{2p}{(1-\rho) + 2p\rho} \rho \right] = W' [\rho^+] \left(\frac{2\rho(1-\rho)}{((1-\rho) + 2p\rho)^2} \right) \quad (82)$$

$$\frac{d}{dp} W \left[\frac{2(1-p)}{(1-\rho) + 2(1-p)\rho} \rho \right] = W' [\rho^-] \left(\frac{-2\rho((1-\rho))}{((1-\rho) + 2(1-p)\rho)^2} \right) \quad (83)$$

$$\frac{d}{dp} \left(\frac{1}{2} (W [\rho^+] + W [\rho^-]) \right)_{p=\frac{1}{2}} = 0 \quad (84)$$

$$\frac{d}{dp} \left(\frac{1}{2} (W [\rho^+] + W [\rho^-]) \right)_{p=1} = W' [\rho^+] \left(\frac{\rho(1-\rho)}{(1+\rho)^2} \right) - W' [\rho^-] \left(\frac{\rho}{1-\rho} \right) \xrightarrow{\rho \rightarrow 1} -\infty \quad (85)$$

So $V_{0D} < V_N$ for p close to 1 as ρ goes to one. In other words, deceiving does not take place in equilibrium as ρ gets close to 1 for very accurate signals.

Step four: Therefore the equation $V_{0D}(q; \rho, W) = V_N(\rho, W)$ has a unique solution $q(\rho, W)$, which maps $[0, 1] \times C_+$ into $[0, 1]$ continuously. The function $V_1(\rho, W) \equiv V_N(\rho, W) = V_{0D}(q(\rho, W); \rho, W)$ defined on $[0, 1] \times C_+$, is nondecreasing in each of its two arguments (must prove this).

A higher q (lower π) makes ρ^+ lower and ρ^- higher (i.e. deceiving more often means less updating in the sense of distance between ρ^+ and ρ^-). As f approaches zero, $V_{0D}(1; \rho, W)$ goes to $\delta W(\rho) = V_{0N}(\rho, W)$. Since V_{0D} is decreasing, this implies q goes to 1 as f approaches 0. Can we likewise prove q goes to 0 as f approaches 1? As f goes to one and p near 1, $V_{0D}(1)$ goes to $\lambda(1-\alpha)^2 \rho^2 + \frac{\delta}{2}(W[\rho^+] + W[0])$. $W(0) = \frac{\lambda\alpha^2 f}{1-\delta}$.

Step five: $\frac{dV_D}{d\rho} \geq \frac{dV_N}{d\rho}$. Combined with the fact that changes in q only affect V_D , this is sufficient for proving $q(\rho, W)$ is nondecreasing in ρ . ■

Proof of Proposition 5. By Proposition 4, T maps C_+ continuously into itself, and is nondecreasing in W . Furthermore, by construction, $T(W + c) = T(W) + \delta c$, for any constant c . By Blackwell's theorem, T is a contraction mapping, and since C_+ with the sup norm is complete, it has a unique fixed point W . Since $q(\rho) = q^*(\rho, W(\rho))$, where $W \in C_+$ is the equilibrium value function, q is continuous in ρ by Proposition 4. ■

Proof of Proposition 6. The initial part of the proof proceeds along the same lines as Theorem 2 of Benabou and Laroque (1992). To show convergence becomes faster for genuine types, we show that $p(\rho^+ - \rho) + (1-p)(\rho^- - \rho) > -(\frac{1}{2}(\rho^+ - \rho) + \frac{1}{2}(\rho^- - \rho))$ for ρ close to 1. First, note that $\rho^+ - \rho = \frac{f(2p-1) - \rho f(2p-1)}{f + (1-f)q + \rho f(2p-1)}$ and $\rho - \rho^- = \frac{f(2p-1) - \rho f(2p-1)}{f + (1-f)q - \rho f(2p-1)}$. Then after cancelling out like terms, the LHS of the inequality becomes

$$p(f + (1-f)q - \rho f(2p-1)) + (1-p)(f + (1-f)q + \rho f(2p-1)) \quad (86)$$

and the RHS becomes

$$-\left(\frac{1}{2}(f + (1-f)q - \rho f(2p-1)) + \frac{1}{2}(f + (1-f)q + \rho f(2p-1))\right) \quad (87)$$

After simplifying, this yields

$$f + (1-f)q > \frac{1}{2}\rho f(2p-1)^2 \quad (88)$$

Since $q(\rho) \rightarrow \frac{f}{1-f} \left((1-\alpha) \sqrt{\frac{\lambda}{k}} - 1 \right)$ as $\rho \rightarrow 1$ (Lemma 1), $\rho \leq 1$, and $2p-1 \leq 1$, this condition simplifies to $\lambda > \frac{k}{4}$. ■

Proof of Proposition 7. Given that $k < \lambda(1-\alpha)^2$, then for f low enough, $\lim_{\rho \rightarrow 1} q(\rho)$ is in the interior of the unit interval (see Lemma 1). Now evaluate limits of value functions as $\rho \rightarrow 1$. $V_D(1) = V_N(1) = \delta W(1)$

(required for $q(\rho) \in (0, 1)$), and $V_H(\rho) = \lambda(2p-1)^2(\alpha + (1-\alpha)\pi)^2 - k + \delta W(1)$. Thus $\lim_{\rho \rightarrow 1} W(\rho) =$

$$\frac{1}{1-\delta} \left(f\lambda(2p-1)^2 \left(\alpha^2 + \frac{(1-\alpha)}{f \left((2p-1)(1-\alpha) \sqrt{\frac{\lambda}{k}} \right)^2} \left(1 - \alpha + 2\alpha f(2p-1)(1-\alpha) \sqrt{\frac{\lambda}{k}} \right) \right) - k \right) \quad (89)$$

We have defined $\bar{W}$ to be the limit of this expression as $f \rightarrow 0$ and $\delta \rightarrow 1$ holding $\frac{f}{1-\delta}$ constant. Thus $\bar{W} =$

$$\frac{f}{1-\delta} \left(\lambda(2p-1)^2 \alpha^2 + 2\alpha(2p-1) \sqrt{\lambda k} \right) \quad (90)$$

This expression is clearly increasing in k , λ , α , p , and $\frac{f}{1-\delta}$. ■

Proof of Proposition 8. Based on numerical results. ■

Proof of Proposition 9. Liquidity conditions are significantly better after the announcement (due to more trades from the passive investor having arrived by this point). Thus the leader is willing to take on a much larger position at this point, and takes on only a reduced position during the pre-announcement trading. Hence he chooses to add to his position more often than he reduces his position. In other words, he expects to buy more if he is not liquidated early. Thus he will not want to disclose unless the expected benefit from disclosure outweighs the expected cost. The statement of the proposition follows from the fact that (1) this expected benefit depends on ρ but the expected cost does not, and (2) expected cost goes to zero as α goes to zero but the expected benefit does not depend on α . ■

Table 1: OLS Estimates of the Effect of Voluntary Disclosure on Stock Returns

	(1)	(2)	(3)	(4)
NEWBUY	0.033**	0.023**	0.037**	0.024*
	(8.38)	(3.81)	(4.97)	(2.41)
LEXISFUND2 x NEWBUY	0.009	0.026**	0.010	0.029*
	(1.53)	(3.02)	(1.29)	(2.56)
LEXISFUND3 x NEWBUY	0.063**	0.117**	0.079**	0.133**
	(7.88)	(10.02)	(4.42)	(6.10)

This table presents the estimated coefficients in a regression of BCAR on NEWBUY crossed with terciles (3 quantiles) of LEXISFUND and including FUNDSIZE quintile dummies. Columns 1 and 3 are estimates for the entire sample, and columns 2 and 4 are estimates for a subsample of firms with market capitalization less than \$20 billion. Columns 1 and 2 are pooled regressions with additional QUARTER dummies and columns 3 and 4 are Fama-MacBeth quarter-by-quarter regressions. There are 54,210 observations in the full sample and 24,494 observations in the restricted sample. T-statistics are below estimates in parentheses, and estimates with t-statistic less than 1 are left as dashes. * denotes significant at 5% ** denotes significant at 1%

Table 2: Two Stage Estimates of the Effect of Voluntary Disclosure on Stock Returns

	(1)	(2)	(3)	(4)
NEWBUY	0.023**	-	0.027**	-
	(4.47)	-	(3.14)	-
LEXRESID2 x NEWBUY	0.016*	0.040**	0.015*	0.039*
	(2.45)	(3.77)	(2.59)	(2.68)
LEXRESID3 x NEWBUY	0.058**	0.101**	0.063**	0.111**
	(8.08)	(9.02)	(5.66)	(6.00)

This table presents the estimated coefficients in a regression of BCAR on NEWBUY crossed with terciles (3 quantiles) of LEXRESID and including FUNDSIZE quintile dummies. Columns 1 and 3 are estimates for the entire sample, and columns 2 and 4 are estimates for a subsample of firms with market capitalization less than \$20 billion. Columns 1 and 2 are pooled regressions with additional QUARTER dummies and columns 3 and 4 are Fama-MacBeth quarter-by-quarter regressions. There are 54,210 observations in the full sample and 24,494 observations in the restricted sample. T-statistics are below estimates in parentheses, and estimates with t-statistic less than 1 are marked as dashes. * denotes significant at 5% ** denotes significant at 1%

Table 3: OLS Estimates of the Effect of Reputation on Stock Returns

	(1)	(2)
NEWBUY	0.005*	0.005
	(2.78)	(1.32)
PERF	0.003	-
	(1.20)	-
NEWBUY x PERF	0.020**	0.042**
	(2.54)	(2.86)

This table presents the estimated coefficients in a regression of FCAR on NEWBUY, PERF, and NEWBUY crossed with PERF and including FUNDSIZE quintile dummies. The sample is restricted to firms with market capitalization less than \$700 million and fund size (by assets under management) in at least the second quintile. Column 1 is a pooled regression with additional QUARTER dummies and column 2 is a Fama-MacBeth quarter-by-quarter regressions. There are 89,123 observations in the sample. T-statistics are below estimates in parentheses, and estimates with t-statistic less than 1 are marked as dashes. * denotes significant at 5% ** denotes significant at 1%

Figure 1: Illustration of timing in the short-run model.

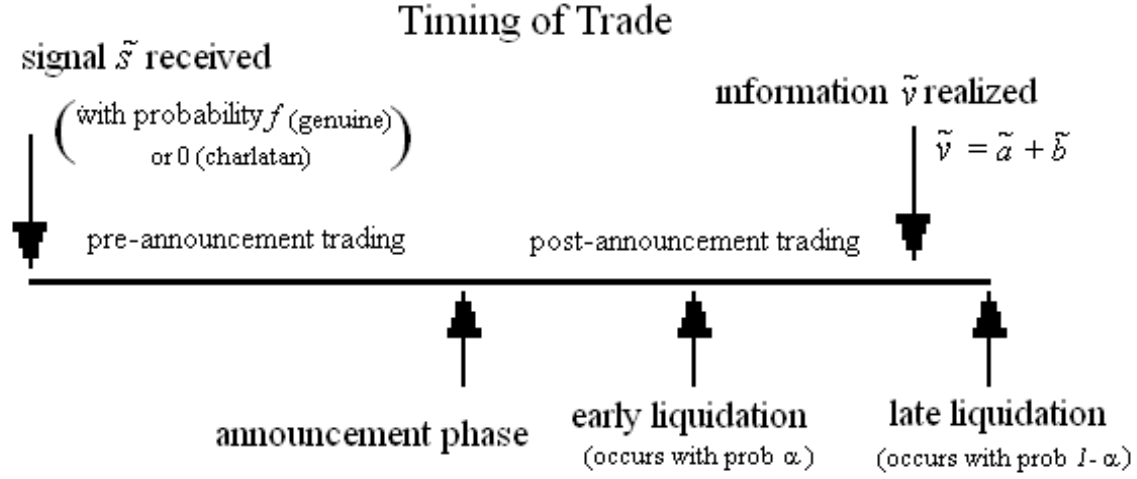


Figure 2: Illustration of timing in the dynamic model.

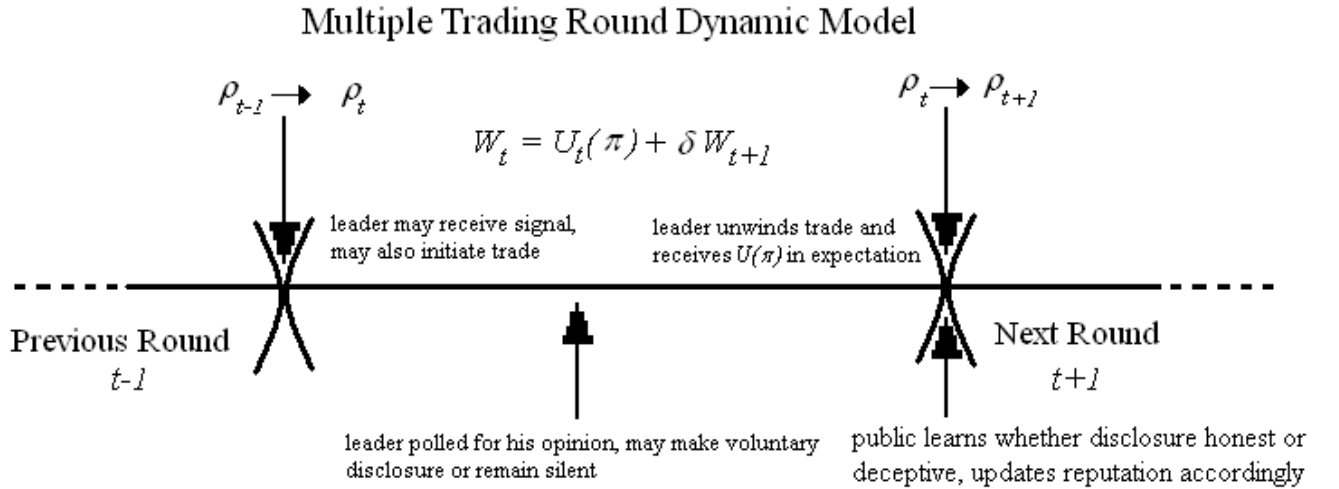


Figure 3: Optimal $q(\rho)$ is at $V_D(\rho, q(\rho)) = V_N(\rho, q(\rho))$. Figure shows V_D and V_N for ρ_1 and ρ_2 , $\rho_1 < \rho_2$.

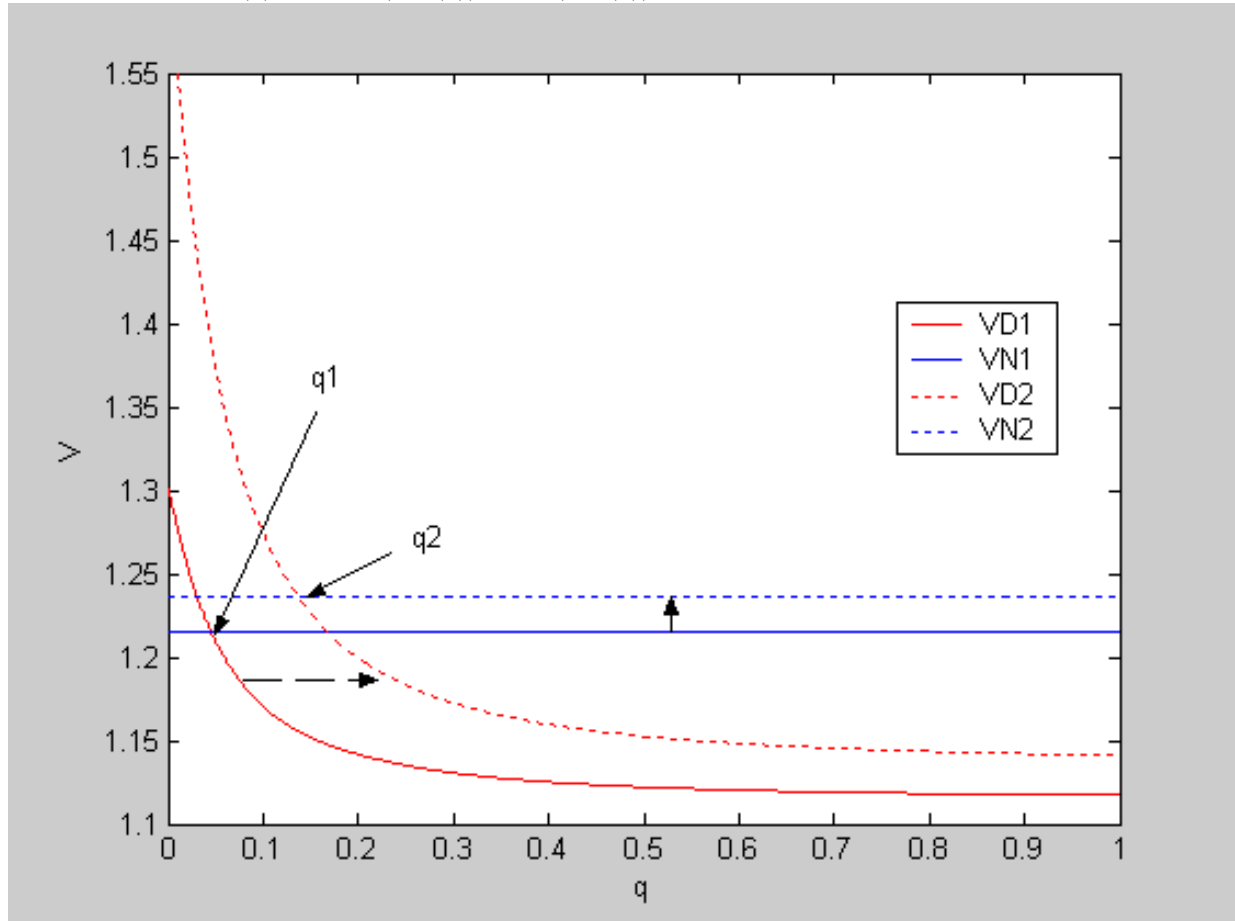


Figure 4: Optimal deception frequency $q(\rho)$.

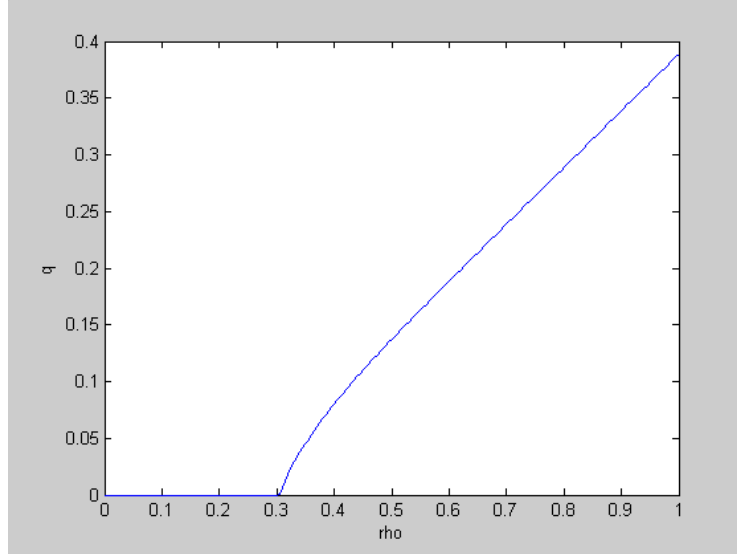


Figure 5: Equilibrium discounted trading profits W .

